ARMMS-T2D

Alliance of Randomized Trials of Medicine vs. Metabolic Surgery in Type 2 Diabetes

A Prospective Consortium Evaluating the Long-term Follow-up of Patients with Type 2 Diabetes Enrolled in a Randomized Controlled Trial Comparing Bariatric Surgery Versus Medical Management

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PRINCIPAL INVESTIGATOR SIGNATURE PAGE

By my signature below, I attest that I have carefully read the attached protocol entitled "Alliance of Randomized Trials of Medicine vs. Metabolic Surgery in Type 2 Diabetes (ARMMS-T2D)", dated 07 October 201, and agree to conduct this study as outlined herein. I will not initiate this study without approval from the appropriate Institutional Review Board (IRB) and understand that any changes in the protocol must be approved in writing by the IRB before they can be implemented.

Principal Investigator: ______________________________ Date ______________

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Contents

ABBREVIATIONS

ADA	American Diabetes Association
AE	Adverse Event
BMI	Body Mass Index
BP	Blood Pressure
C Peptide	Connecting Peptide
CBC	Complete Blood Count
CI	Confidence Interval
CRF	Case Report Form
CVD	Cardiovascular Disease
DCC	Data Coordinating Center
DCCT	The Diabetes Control and Complications Trial
OSMB	Observational Study Monitoring Board
eGFR	Estimated Glomerular Filtration Rate
EMS	Emergency Medical Support
ESRD	End Stage Renal Disease
EWL	Excess body Weight Loss
FDA	Food and Drug Administration
GLP	Glucagon-Like Peptide
HbA_{1c}	Hemoglobin A1c
HIPAA	Health Insurance Portability and Accountability Act
hsCRP	High sensitivity C Reactive Protein
ICF	Informed Consent Form
IRB	Institutional Review Board
ITT	Intention-To-Treat
LABS	Longitudinal Assessment of Bariatric Surgery
LAGB	Laparoscopic Adjustable Gastric Banding
LSG	Laparoscopic Sleeve Gastrectomy
NGSP	National Glycohemoglobin Standardization Program
NHLBI	National Heart, Lung and Blood Institute
PHI	Protected Health Information
RCT	Randomized Control Trial
RYGB	Roux-en-Y Gastric Bypass
SAE	Serious Adverse Event
SAP	Statistical Analysis Plan
SOS	Swedish Obese Subjects Study
T2D	Type 2 Diabetes
TZD	Thiazolidinediones
UKPDS	The United Kingdom Prospective Diabetes Study

1.0 STUDY AIMS

1.1 Specific Aims

Aim 1 – To compare the durability of glycemic control, assessed by HbA_{1c}, between patients randomized to bariatric/metabolic surgery or comprehensive medical/lifestyle therapy.

For our primary outcome, we will examine between-group differences in the percent change of HbA_{1c} over time from baseline to 7 years for all participants, and up to 13 years for the earliest enrollees. We *hypothesize* that, among persons with T2D and a baseline BMI of 27-45 kg/m^2, the percent change in HbA_{1c} will be clinically and significantly greater following random treatment allocation to bariatric/metabolic surgery compared to medical/lifestyle therapy. We will also compare the following clinically important secondary outcomes across treatment groups: a) rate of glycemic control, b) rate of diabetes remission, and c) rate of diabetes relapse after initial remission.

Aim 2 – To assess the longer-term efficacy and safety outcomes in patients randomized to bariatric/metabolic surgery or comprehensive medical/lifestyle therapy.

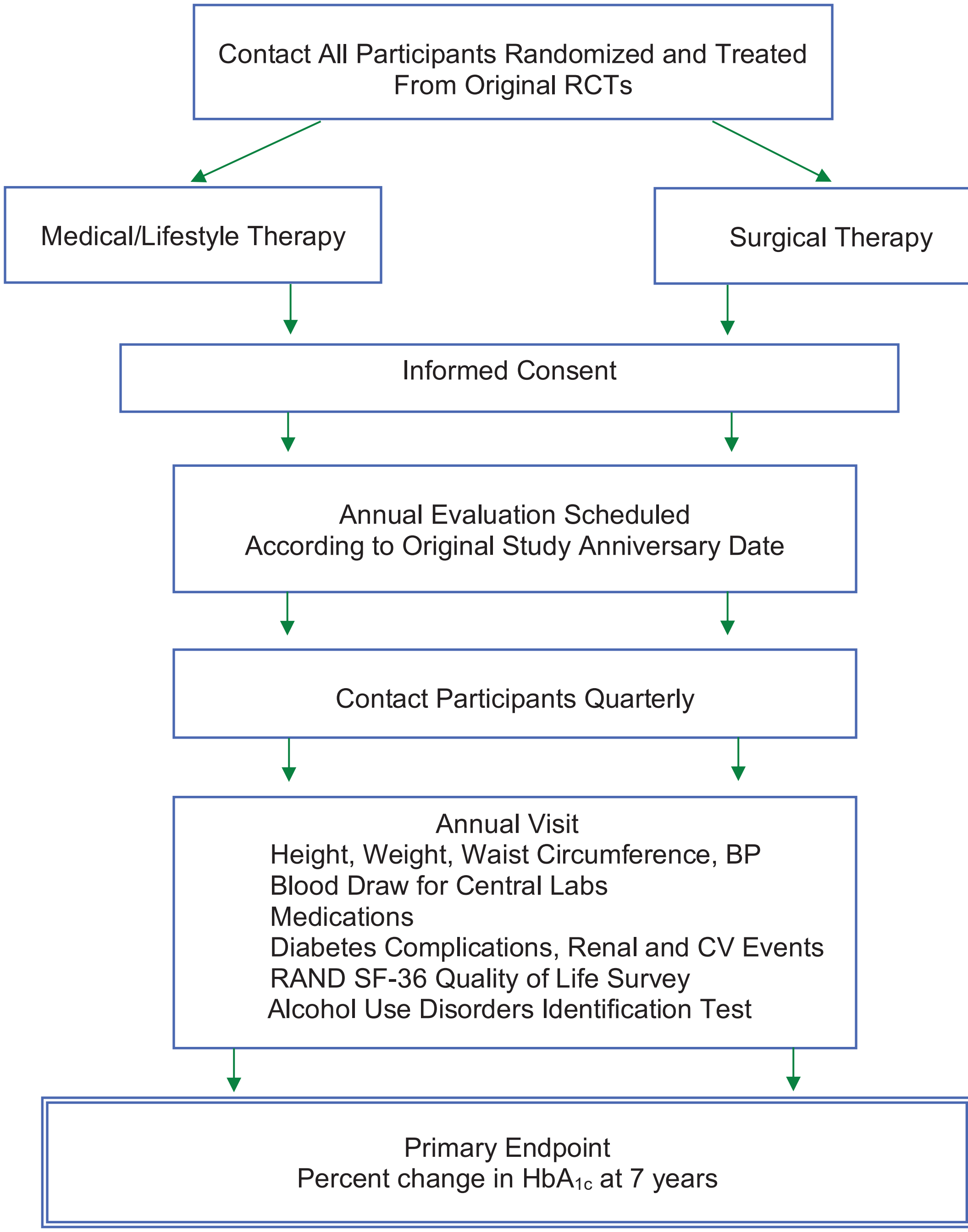
For *efficacy*, we will test the *hypothesis* that the percent change over time in weight/BMI, cardiometabolic risk factors (lipids, blood pressure, albuminuria), cardiometabolic risk score, and quality of life, will be greater for those randomized to bariatric/metabolic surgery compared to medical/lifestyle therapy from baseline to 7 years follow-up for all participants, and up to 13 years for the earliest enrollees. For our *safety outcomes*, we will report major adverse events (e.g., hospitalizations, deaths, serious hypoglycemia).

Aim 3 – To identify clinical predictors of diabetes remission and relapse.

We *hypothesize* that baseline age, T2D duration, pre-intervention insulin use, HbA_{1c}, glucose and insulin levels, procedure type, and amount of initial weight loss will be significant predictors of long-term diabetes remission and relapse.

Impact - The ARMMS-T2D study will provide the largest body of long-term Level-1 evidence to inform clinical decision-making regarding the comparative durability, efficacy, and safety of bariatric/metabolic surgery relative to medical/lifestyle care of diabetes, including among patients with only mild obesity or mere overweight.

1.2 Study Flow Diagram



2.0 INTRODUCTION

2.1 Background

Type 2 diabetes (T2D) and obesity are now endemic worldwide, causing significant morbidity related to renal and cardiovascular diseases, and are the driving force behind preventable mortality. Recent national surveys show that 29.1 million Americans, or 9.3% of the adult population, have diabetes and 34% are obese, which together contribute to substantial individual health burden and societal healthcare costs. The short-term safety and efficacy of currently practiced bariatric procedures to improve metabolic control and induce initial remission of T2D in severe obesity (BMI ≥35 kg/m^2) is established, and emerging data indicates that bariatric surgery is also effective in patients with BMI <35 kg/m^2. Long-term observational studies suggest that bariatric surgical procedures improve survival and reduce cardiovascular events among patients with diabetes; however, observational studies are limited by considerable unmeasured selection bias. In addition, advances in pharmacologic and multidisciplinary lifestyle approaches to diabetes management have led some to question whether bariatric surgery should be considered as a therapeutic approach to diabetes management.

2.2 Rationale for Aim 1

The NIH sponsored several randomized control trials (RCTs) to evaluate the efficacy of bariatric/metabolic surgery compared to multidisciplinary medical and lifestyle management of T2D and body weight. These trials assessed feasibility, with a relatively short follow-up of 1-3 years, and showed significantly greater diabetes remission in the surgery compared to medical therapy groups. However, they were not of sufficient duration to provide long-term data to assess durability, and hence the data that would support recommendations for the use of bariatric/metabolic surgery to treat lower-BMI patients with T2D. To address the need for long-term RCT data, investigators from four of the NIH funded RCTs (*Cleveland Clinic, Joslin Diabetes Center/Brigham and Women's Hospital (BWH), University of Pittsburgh,* and the *University of Washington*) formed the Alliance of Randomized Trials of Medicine versus Metabolic Surgery in Type 2 Diabetes (***ARMMS-T2D***) consortium. These RCTs include a set of common measures, and all completed at least two years of follow-up. When combined, participants from these studies represent the largest cohort of patients with T2D (35% having a BMI <35 kg/m^2) ever to have undergone randomized assignment to bariatric/metabolic surgical procedures or medical/lifestyle intervention. Across the four studies, 355 patients were originally randomized, and 302 patients are currently available for long term follow-up. This rare and valuable cohort provides a unique, timely, and cost-effective opportunity to obtain essential data on the durability of bariatric surgery effects and changes-over-time pertaining to diabetes and related conditions.

Diabetes and Bariatric Surgery

People with diabetes can reduce the occurrence of micro- and macro-vascular complications by controlling blood glucose, lipids, and blood pressure. However, the intensification of medical therapy that is required to sufficiently reduce levels of glycosylated hemoglobin (HbA_{1c}) can result in weight gain and an increased risk of hypoglycemia. These two factors - weight gain and the fear of hypoglycemia - often limit patients' potential to reach euglycemia. Despite the widespread availability of insulin and oral hypoglycemic agents, the prevalence of T2D has rapidly increased over the past two decades. Furthermore, the best available medical and lifestyle treatment for T2D has failed to result in a reduction in long-term cardiovascular risk **(1)**.

Among severely obese patients, bariatric surgery causes substantial, sustained weight loss. Bariatric operations range from gastric-restrictive, to malabsorptive, to gastro-entero-neuronal-hormonal altering procedures, including laparoscopic adjustable gastric banding (LAGB), laparoscopic sleeve gastrectomy (LSG), Roux-en-Y gastric bypass (RYGB), and the biliopancreatic diversion. The three dominant procedures used today are RYGB, LSG, and LAGB, which together account for at least 95% of all bariatric operations performed in the United States.

In non-randomized observational studies, bariatric surgery is associated with improvements in virtually all obesity-related co-morbidities as compared to non-surgical treatment. This is especially notable for treatment, prevention and control of diabetes. Moreover, observational studies suggest that bariatric surgery improves long-term survival **(2-8)**. However, in the absence of randomized clinical trials, it is possible that these conclusions are biased because healthier and/or more motivated patients may have opted or been selected for surgical intervention, providing substantial potential for unmeasured selection bias. It is this potential confounding in observational studies that most underscores the need for prospective randomized controlled studies of bariatric surgery.

"Metabolic Surgery" for Diabetes

In 1995, Pories and colleagues published a seminal study on the effectiveness of bariatric surgery for the treatment of T2D in obese patients **(9)**. Subsequently, the 1st Diabetes Surgery Summit (DSS-I) convened in Rome in 2007, reviewed the available clinical and mechanistic data of the time, and concluded that gastrointestinal surgery, or "metabolic surgery", could be used to treat T2D in patients with only mild obesity **(10)**. These recommendations were reinforced at the 2nd Diabetes Surgery Summit (DSS-II), held in London in 2015, where a multidisciplinary international group used the Delphi process to conduct an evidence appraisal of metabolic surgery as a treatment for T2D. DSS-II concluded that *metabolic surgery* should be *recommended* to treat T2D in patients with a BMI $\geq$40 kg/m^2 and in those with a BMI of 35.0-39.9 kg/m^2 when hyperglycemia is inadequately controlled by lifestyle and medical therapy, and it should be *considered* for those with a BMI of 30.0-34.9 kg/m^2 if glycemic control cannot be achieved by diabetes medications **(11)**. However, DSS-II also recognized that one of the greatest limitations to recognizing and adopting these recommendations was the absence of data from RCTs demonstrating the long-term efficacy, safety, and durability of metabolic surgery, especially in patients with only mild obesity (BMI <35 kg/m^2). ARMMS-T2D represents the best currently available option to fill this unmet evidence need.

Metabolic surgery involving some degree of intestinal bypass appears to exert the most dramatic effects on diabetes. Approximately 70% of people with severe obesity and diabetes who undergo RYGB experience disease remission, maintaining glycemic control without medications for many years **(3, 7, 9, 12-17)**. Data for sleeve gastrectomy and gastric banding in obese patients with diabetes **(18)** are less clear. Other than data from the STAMPEDE study (one of the studies that comprises ARMMS-T2D), there are no RCTs with longer-term glycemic control data for LSG. For LAGB, there are no longer-term RCT data that can replicate the results achieved outside the U.S. for glycemic control **(19)**. Therefore, understanding the procedure-specific differences compared to non-surgical treatment is also an important goal, and this would help inform procedure choice for patients. For diabetes, which is traditionally viewed as a chronic, progressive disease in which delay of end-organ complications is the major treatment goal, metabolic surgery offers a novel possibility: the concept of durable disease remission. Although relapses do occur with long-term follow-up **(14)**, early and prolonged metabolic improvements may still reduce diabetes-related complications, with differences increasing over time **(20)**, and with reduced need for anti-diabetes medications.

There is now strong evidence that diabetes resolves early after RYGB, before major weight loss has occurred **(9, 21)**. Emerging data demonstrates that these anti-diabetes effects are mediated not only by reductions in food intake and body weight, but also by weight-independent mechanisms **(13, 22-37)**. Consistent with this concept, preliminary non-randomized observational studies suggest the anti-diabetes impact of RYGB is approximately as great among less obese persons (e.g., BMI <35 kg/m^2), who lose less weight postoperatively compared to the severely obese **(13, 38-42)**. Similarly, numerous studies indicate RYGB, and experimental variations of the operation, reverse diabetes in both obese and non-obese diabetic animals **(13, 43-48)**. Therefore, the concept of "metabolic surgery" targeted primarily to treat diabetes, rather than obesity, is increasingly recognized as an acceptable therapeutic strategy for diabetes management **(11, 13, 49, 50)**. However, many questions remain regarding the proper role of surgery in overall diabetes care, and there are no RCTs related to long-term outcomes including risks and durability to help guide clinical practice.

2.3 Review of Prior Studies

Despite a tremendous increase in the number of bariatric/metabolic operations performed in the United States and worldwide, there is a gap between the proliferation in usage of these procedures and the quality of the long-term evidence needed to inform clinical decisions regarding their appropriate use, and the durability of outcomes. This research gap includes lack of randomized treatment assignment, with structured assessment of the *long-term* effectiveness and risks of bariatric surgery compared to medical management for patients with diabetes.

Existing long-term data on bariatric surgery for patients with diabetes include the following non-randomized observational studies.

- *Swedish Obese Subjects Study (SOS)* **(5, 6, 8, 51)**: A prospective non-randomized matched cohort analysis of bariatric surgery and usual care. It includes only 12% gastric bypass patients (who came into the cohort latest), had no LAGB or LSG procedures, and involved only 18% of patients with diabetes at baseline. This study did not match on motivation, personality, or other important characteristics that may be important in comparing outcomes across patients who self-selected surgical vs. non-surgical treatment. A major concern regarding all of its findings is that because the participants who had surgery actively chose to undergo this aggressive intervention, perhaps they were more motivated overall to improve their health, and therefore they may also have been more compliant with medications, led healthier lifestyles, etc. These potential confounders limit the generalizability of all SOS reports and all other non-randomized observational studies of bariatric surgery listed below.
- *The Longitudinal Assessment of Bariatric Surgery Consortium (LABS)* ***(52, 53)**:* A non-randomized, multi-institutional, prospective observational evaluation of bariatric surgery. The domains of interest include safety, longitudinal efficacy evaluations, and prospective assessment of a range of additional outcomes, including mechanistic explorations. Importantly, the LABS trial does not include non-surgical comparators, and only 33% of participants had diabetes at baseline **(53)**.
- *Cleveland Clinic Study on Surgical Cure for Diabetes* ***(54)**:* A retrospective analysis of diabetes outcomes ≥5 years after bariatric surgery in 217 patients. There was a complete remission rate (HbA_{1c} <6%, fasting blood glucose <100 mg/dL) of 24%, a partial remission rate in an additional 26% (HbA_{1c} = 6.0-6.4%, fasting blood glucose = 100-125 mg/dL), and improvement in a further 34% of patients. 19% of patients had long-term recurrence of diabetes after achieving remission. Longer duration of T2D, and lower long-term excess weight loss (EWL) predicted lack of complete remission and recurrence. This study used precise definitions of remission and is the only study including sleeve gastrectomy.
- *Utah Obesity Study (2, 55, 56):* A matched analysis in a cohort of patients who underwent bariatric surgery at one center. Because this was also not a randomized study, it is plagued by similar concerns regarding unmeasured confounding as described for SOS.

- *VA Comparative Study* ***(57)****:* A retrospective cohort study of patients who underwent bariatric surgery at VA medical centers throughout the United States, compared with nonsurgical controls at similar centers. Surprisingly, unlike the above-mentioned SOS and Utah studies, surgery was not associated with decreased mortality over 7 years in propensity score-adjusted analyses. These results, however, were derived from predominantly older male patients with high baseline mortality rates – a skewed population from a unique practice environment; patients were very different from the younger, primarily female population that typically chooses bariatric surgery. The findings, therefore, are difficult to generalize, and importantly for the current application, the work did not focus on diabetes nor include randomized participants.
- *HMO Research Network Studies:* A large network of integrated health insurance plans and care delivery systems is conducting a series of retrospective observational studies of bariatric surgical procedures in patients with and without diabetes **(14, 58, 59)**. To date, these studies have suggested that bariatric surgical procedures are superior to non-surgical treatment of diabetes in inducing long-term diabetes remission events, but these studies have several important limitations, including the lack of an intensive medical/lifestyle intervention control group and potential for unmeasured confounding in the control non-surgical group (similar to SOS as noted above) related to motivation, personality, or other important characteristics that may be important in comparing outcomes.

Notably, very few of the subjects in prior observational studies were moderately obese (BMI 30-35). The Diabetes Surgery Symposium held in Rome, Italy in 2007, comprised of an international group of experts across multiple disciplines, concluded with strong consensus that "gastrointestinal surgery may be appropriate for the treatment of T2D in patients who are good surgical candidates with a BMI of 30–35 kg/m^2 and are inadequately controlled by lifestyle and medical therapy" **(10)**. However, despite these recommendations and best judgment, the role of an operative intervention for diabetes treatment among moderately obese patients has not yet been sufficiently addressed by available research.

Randomized controlled trials

Recent systematic reviews summarized all RCTs that have compared bariatric surgery against non-surgical treatments for obesity **(60, 61)**. There are currently 11 RCTs comparing bariatric/metabolic surgery with medical/lifestyle treatment of T2D, including four from the ARMMS-T2D consortium **(60, 62-68)**. In total, these studies randomized 794 patients with T2D, and they vary in size from 38 to 150 participants. Follow-up has ranged from 6 months to 3 years, with one of the smaller RCTs recently reporting out to 5 years **(16)**. These studies generally focused on cohorts with T2D with 1 to 2 year follow up, and they provided good evidence that bariatric procedures, including RYGB **(63, 69, 70)**, LAGB **(71, 72)**, BPD **(70)**, and LSG **(63)**, result in greater short-term (1-2 years) weight loss (mean difference -26 kg; 95% CI: -31 to -21; P<0.001) and greater remission of T2D [complete case analysis Relative Risk (RR) of T2D remission: 22.1; 95% CI: 3.2 to 154.3; P=0.002; conservative analysis RR: 5.3; 95% CI: 1.8 to 15.8; P=0.003] compared to a variety of non-surgical treatment options **(63, 69-72)**. Two additional small, randomized controlled trials have been published showing similar short-term results for both weight loss and T2D **(65, 67)**. In addition, serum triglycerides and high density lipoproteins were more favorably affected by bariatric procedures, but blood pressure and other lipoproteins were not (although some of the studies did show reductions in medication use for these conditions) **(61)**. These reviews also noted that there is a lack of sufficient evidence from RCTs beyond 2 years, specifically with respect to durability of diabetes control, mortality, CVD, and adverse events **(61)**. Another recent systematic review focused on weight loss and glycemic control in class I obese (BMI 30-34.9 kg/m^2) adults with T2D and identified 3 RCTs showing results that were similar to those seen in class II (BMI 35-39.9 kg/m^2) and severely obese populations; however, the review also noted a lack of long-term studies **(73)**. Notably, two of the longest RCTs to date include the 3-year STAMPEDE **(62)** and TRIABETES trials **(64)**, and are included in this application. Further, 5-year data from the STAMPEDE trial also suggest sustained benefit of surgery in mild to moderately obese patients with T2D. The data demonstrate the sustained superiority of surgery versus intensive medical/lifestyle therapy in moderate to severely obese patients with T2D (74). Our current application will include the participants from this latter trial, and is poised to provide the durability data that will bridge the gap in our current understanding of long-term effects of surgery for the management of T2D.

A randomized controlled study by Dixon et al. demonstrated that the reduction in diabetes risk following LAGB compared to a non-surgical treatment group of patients with BMI of 30-40 kg/m^2 was due to the amount of weight loss after that operation **(71)**. The medical weight loss program in this study resulted in <2 kg weight loss at one year. An earlier study by the same group reported greater improvement in metabolic syndrome in those patients with a BMI of 30-40 kg/m^2 randomized to an intensive medical management program, but the improvements were still less than those in the LAGB group **(7)**. In contrast, the multi-center Look AHEAD study has shown an 8.6% weight loss and reduction in HbA_{1c} from 7.3% to 6.6% at 12 months in response to an intensive lifestyle intervention. One-year data from the three reported randomized trials conducted at the Cleveland Clinic, Boston, and Pittsburgh sites of our ARMMS-T2D Consortium demonstrate that bariatric surgical approaches provide improved glycemia, lipid profiles, weight and blood pressure, compared to medical/lifestyle approaches that were all of higher intensity than standard medical care. The Cleveland Clinic group recently published 5-year follow-up data for the STAMPEDE cohort that support the durability of surgery in maintaining glycemic goals in more patients than those treated with medical therapy (62, 74). While these data are encouraging, there is still a major gap in our knowledge regarding the long-term (>5 years) durability and safety of these procedures.

2.4 Rationale for Aims 2 and 3

Recent national estimates suggest that 29.1 million Americans, or 9.3% of the population, have diabetes (75), and 34% of the adult population is obese (76). Together, type 2 diabetes (T2D) and obesity contribute to substantial individual health burden and societal healthcare costs. In the United States, diabetes is the number two cause of hospitalizations in adults aged 18 years or older, accounting for approximately 11% of admissions. With no medical cure, the natural course of diabetes is characterized by progressive beta cell failure and can be accompanied by development of microvascular and macrovascular complications, leading to renal failure, blindness, amputation, and death due to cardiovascular disease (CVD). Bariatric surgery has emerged as an effective clinical tool for weight loss and initial control of T2D (63, 69-71).

Diabetes Outcomes and Intensive Medical Treatments

Persisting Unknowns Regarding Optimal Medical & Lifestyle T2D Management. Concurrent with an expanding experience with bariatric surgery, there have also been significant advances in non-surgical treatments of diabetes. This has led some to question why surgical options should be considered as primary therapy at the very time when medical, non-surgical treatments show improved efficacy **(77)**. Since 1995, multiple diabetes drugs have been approved by the FDA, including metformin, alpha-glucosidase inhibitors, thiazolidinediones (TZDs), glinides, GLP-1 analogs, amylin analogs, DPP-IV inhibitors, a bile acid sequestrant, a dopamine agonist, and SGLT-2 inhibitors. Among these, two new agents, empagliflozin and liraglutide, have been shown to improve major adverse cardiovascular event rates **(78, 79)**. In addition, multiple insulin analogs with improved kinetics and safer dosing profiles are now available. Together, these agents hold great promise to generate improved health outcomes for patients with diabetes, and NHANES data demonstrate that more patients are achieving blood sugar, blood pressure, and lipid goals **(80)**. Nevertheless, fewer than 20% of patients with diabetes are meeting all three health goals simultaneously. Long-term safety of pharmacological agents remains incompletely understood, and some may impart worrisome adverse outcomes, such as weight gain and hypoglycemia with many diabetes medications, possible pancreatitis with GLP-1 analogues, and various cancers with insulin analogues, TZDs, and/or SGLT-2 inhibitors. In addition, very tight glycemic control might not reduce cardiovascular event rates **(81-83)**, and may even lead to increased mortality **(83)**. Thus, optimal treatment strategies and glycemic goals for patients with diabetes remain incompletely understood. Similarly, lifestyle modification (primarily diet and exercise) can promote successful weight loss, improving glycemia and complications in patients with diabetes **(84)**. Multidisciplinary weight management approaches are emerging as viable and potentially cost-effective treatment options in overweight or obese patients with diabetes **(10, 33, 85)**. Such approaches, however, often fail to achieve durable weight loss of more than 5-10%. Indeed, over time participants in the SOS study who received standard medical treatment for obesity realized an average weight *gain* of 1.6% over 10 years **(86)**. Importantly, these lifestyle and medical approaches do not appear to improve cardiovascular outcomes over a follow-up period up to 13.5 years **(87)**. Thus, further research is needed to evaluate current medical and lifestyle therapeutic regimens for diabetes, and how such approaches compare to surgical interventions.

In summary, there still remain many unanswered questions about the relative clinical utility of surgical compared with medical treatment of diabetes and obesity/overweight, especially in the lower BMI range. Addressing these questions will require long-term follow-up of randomized controlled studies to more completely inform health care decision-making and clinical practice. Importantly, it is not clear how well diabetes outcomes are influenced over the long term by the type of surgery or by the amount of weight lost, or if bariatric surgery is more effective than non-surgical weight loss induced by diet and physical activity and glucose lowering medications in reducing long-term micro- and macrovascular complications among diabetic patients with more moderate BMIs (e.g., Class I and Class II obesity). This study will address, in a systematic way, these important and yet unanswered questions.

It is important to consider the correct comparator to bariatric surgery. It is not appropriate to compare bariatric surgery to standard of care, as currently practiced. In current practice, the majority of patients are not meeting glycemic, lipid and blood pressure goals **(88)**. It may be argued that given the cost of surgical intervention, appropriate medical interventions for comparison should be much more rigorous, and include approaches such as residential treatment for several weeks to initiate rapid weight loss under medical supervision, home-based treatment for several months, with provision of prepared meals consistent with dietary goals and sports or fitness club memberships **(89)**; however the pragmatic application of such comparison would be limited. Thus, each medical management arm of the participating parent trials was based on similar Diabetes Prevention Program **(90)** and Look AHEAD models **(91)**, albeit implemented in different ways with focus ranging from predominantly lifestyle intervention in the TRIABETES trial **(92)** to an intensive pharmacologic plus lifestyle intervention approach in the Stampede trial **(63)**, and between in the Crossroads **(93)** and SLIMM-T2D trials **(94)**. Weight loss at one year ranged between 5-10% and HbA_{1c} ≤6.5% was demonstrated in ~16-35% of the medical management cohorts in our ARMMS-T2D Consortium trials **(62, 63, 65, 67)**, demonstrating that the approaches to medical intervention achieved results superior to those anticipated with standard of care treatment, but using approaches that are considered reasonable in implementation.

Microvascular Events and Bariatric/Metabolic Surgery

Intensive glycemic control (HbA_{1c} <7%) using medications has been shown to decrease the risk of microvascular complications (retinopathy, nephropathy, and neuropathy) associated with diabetes in two landmark randomized medical-therapy trials, both with >10 year follow-up **(95,**

96). The Diabetes Control and Complications Trial (DCCT) monitored the risk of developing microvascular complications in type 1 diabetes mellitus. Compared to patients receiving standard therapy, those randomized to intensive glycemic control demonstrated greater improvement in HbA_{1c} and a reduction in the risk of developing or progression of retinopathy, nephropathy, and neuropathy by 50% or greater.

The United Kingdom Prospective Diabetes Study (UKPDS) provided definitive information on the relationship between improved glycemic control and prevention of complications in T2D. Patients with newly diagnosed diabetes were randomized to intensive *vs*. conventional therapy. Those receiving intensive therapy maintained lower HbA_{1c} values (7.0% compared with 7.9%; P<0.001), which was associated with a significant 25% risk reduction (P=0.009) in combined microvascular endpoints (eye, kidney, and nerve). There was also a tendency toward fewer myocardial infarctions in the group assigned to intensive therapy. In a sub-group of overweight patients assigned to intensive treatment with metformin or conventional treatment, those receiving intensive treatment maintained a 0.6% lower HbA_{1c}, resulting in a 32% lower risk for any diabetes-related endpoint (P=0.002), and a 36% lower risk of death from any cause (P=0.021).

These studies both showed that for every 1% absolute drop in HbA_{1c} (e.g., from 9.0% to 8.0% HbA_{1c}), there was a relative risk reduction for microvascular disease of 25-45%. Glycemic control to achieve an HbA_{1c} of 7.0% or lower in some of these patients was only possible with insulin doses >100 units per day, and even then for only short periods of time **(96)**. Only half of patients with diabetes are currently meeting glycemic goals **(80)**. It is believed that in community settings where 95% of diabetic patients are treated, HbA_{1c} levels in patients with diabetes typically vary from 8.5 to 9.0%, suggesting that control with medications outside of clinical trials is difficult to achieve and maintain. Furthermore, the advanced medical therapy required to achieve an HbA_{1c} of 7.0% confers a corresponding 2-4% yearly incidence of severe hypoglycemia **(96)**.

Improvements in blood pressure and lipids also need to be considered following bariatric surgery and medical management, as these important comorbid conditions contribute to the micro- and macro-vascular complications of diabetes. Moreover, longer term studies of both the DCCT **(97, 98)** and UKPDS **(99)** cohorts demonstrate persistent health benefits of earlier glycemic control, referred to as "metabolic memory", with emergent cardiovascular benefit over a much longer time period even long after intensive glycemic control had ceased **(100, 101)**. Together, these findings demonstrate an urgent need for improved diabetes management options and the potential for long-term health impact following bariatric procedures, given their profound effects to promote diabetes remission or improvement. However, to date there are minimal to no long-term term data on the effects of bariatric surgery on diabetes complications, particularly in patients with lower BMI **(102)**.

Predictors of Diabetes Remission

Data from the ARMMS-T2D parent trials together with related RCT data show that not all patients achieve glycemic control after metabolic surgery without the need for medications, or they are unable to sustain long-term control despite initial success **(7, 9, 12, 13, 16, 54, 62-71)**. There have been some attempts to develop statistical models that would predict the probability of remission, including the DiaRem and ABCD scores **(103-105)**. These models are based on preoperative patient data including age, BMI, C-peptide levels, duration of diabetes, etc. We have reported that adding postoperative weight loss to DiaRem may improve predictability **(106)**. The lack of robust RCT data from a sufficient number of patients and from key demographic groups – e.g., patients with a BMI <35 kg/m^2 – is currently limiting the utility and accuracy of prediction. ARMMS-T2D will use exploratory tools to address this issue under Specific Aim 3.

Consortium of RCTs

Previously, four investigative groups initiated RCT's at their respective site to evaluate the effectiveness of bariatric surgery compared to multidisciplinary medical and lifestyle management of diabetes and body weight. These trials were each designed to assess feasibility with a relatively short duration of follow-up of only 1-3 years. Individually, each trial lacks the sample size and duration of follow-up to meaningfully inform clinical decision making. Together, with longer follow-up, these randomized trials can provide a unique national resource to address timely and unanswered clinical questions related to the durability of these alternative management approaches in patients with T2D and obesity.

Investigators from these four RCTs (Appendix 1) have unified to form the Alliance of Randomized Trials of Medicine versus Metabolic Surgery in Type 2 Diabetes (ARMMS-T2D) Consortium to support a longer-term (7 additional years) follow-up of 302 participants who were randomized to surgical vs. non-surgical diabetes management approaches. These RCTs include a set of common measures, and all have completed a minimum of one-year follow-up assessments (Appendix 2). Together, participants from these studies represent the largest cohort with diabetes (one third having a BMI <35 kg/m^2) ever to undergo randomized assignment to bariatric surgical procedure vs. medical/lifestyle intervention.

3.0 STUDY OBJECTIVES

The aim of the study is to assess the durability and effectiveness of bariatric surgery compared to medical/lifestyle management of T2D in obese patients at seven or more years' follow-up.

Specifically, our primary outcome will examine between-group differences in the percent change of HbA_{1c} over time from baseline to 7 years for all participants, and up to 13 years for the earliest enrollees. We *hypothesize* that, among persons with T2D and a baseline BMI of 27-45 kg/m^2, the percent change in HbA_{1c} will be clinically and significantly greater following random treatment allocation to bariatric/metabolic surgery compared to medical/lifestyle therapy. We will also compare the following clinically important secondary outcomes across treatment groups: a) rate of glycemic control, b) rate of diabetes remission, and c) rate of diabetes relapse after initial remission.

The study also seeks to determine longer-term efficacy and safety outcomes in patients randomized to bariatric/metabolic surgery or comprehensive medical/lifestyle therapy. For *efficacy*, we will test the *hypothesis* that the percent change over time in weight/BMI, cardiometabolic risk factors (lipids, blood pressure, albuminuria), cardiometabolic risk score, and quality of life, will be greater for those randomized to bariatric/metabolic surgery compared to medical/lifestyle therapy from baseline to 7 years of follow-up for all participants, and up to 13 years for the earliest enrollees. For our *safety outcomes*, we will report major adverse events (e.g., hospitalizations, deaths, serious hypoglycemia).

We will also identify clinical predictors of diabetes remission and relapse. We *hypothesize* that baseline age, T2D duration, pre-intervention insulin use, HbA_{1c}, glucose and insulin levels, procedure type, and amount of initial weight loss will be significant predictors of long-term diabetes remission and relapse.

Subject to IRB approval and subject informed consent, this study will include an additional research component involving collection of biological samples (biobank) for de-identified exploratory analysis.

The ARMMS-T2D study will address major gaps in our current knowledge base and provide long-term evidence to inform decision-making regarding the role of bariatric surgery compared to medical/lifestyle management of diabetes among patients with mild to moderate obesity.

4.0 STUDY DESIGN

4.1 General Design

This is a prospective long-term observational follow-up study of participants who underwent bariatric surgery or medical/lifestyle therapy in an RCT at one of the 4 participating sites. Approximately 302 participants from the four RCT sites will be eligible to re-consent. Appendix 2 provides a summary of the initial enrolled participants at the collaborating sites, their demographics and potential available participants in the ARMMS-T2D study. Data common across the four sites will be collected for a total of five years' follow-up.

Study Population

The study population will consist of adult patients with T2D and mild-to-moderate (Class 1 and 2) obesity or higher levels of overweight (BMI range 27-45 kg/m^2). Specifically, the study population will include participants recruited and randomized to either bariatric surgery or a non-surgical medical/lifestyle intervention to treat diabetes and initiated intervention, at one of four sites participating in the ARMMS-T2D Consortium.

Original inclusion criteria for participation in the RCTs at all sites included:

- Candidate for general anesthesia or unsupervised exercise.
- Age ≥20 and ≤65 years.
- Body mass index >27 and ≤45 kg/m^2.
- Diagnosis of type 2 diabetes confirmed by either requiring diabetes medication and/or having elevated glycemia based on HbA_{1c}, fasting plasma glucose, and/or OGTT results, according to American Diabetes Association criteria **(107)**.
- Ability and willingness to participate in the study and agree to any of the research arms.
- Able to understand the options and to comply with the requirements of each program.
- Negative urine pregnancy test at screening and baseline visits (prior to surgery) for women of childbearing potential (i.e., biologically capable of becoming pregnant). This includes women who are using contraceptives or whose sexual partners are either sterile or using contraceptives.

4.2 Primary Study Endpoints

The primary endpoint is the percent change in HbA_{1c} from baseline between the surgical and non-surgical groups

4.3 Secondary Study Endpoints

The following parameters will be analyzed as secondary endpoints. Methods of analysis are fully described in the Statistical Analysis Plan (SAP).

Long Term Diabetes Control

1. Glycemic control: percentage of participants achieving HbA_{1c} < 7.0% on or off diabetes medications.
2. Diabetes remission: percentage of participants achieving HbA_{1c} <6.5% off diabetes medications.
3. Diabetes partial remission: percentage of participants achieving HbA_{1c} <6.5% with or without diabetes medications
4. Diabetes relapse: time from the initial remission to relapse (only defined for participants with remission)

Body Composition

1. Percent change of body weight from baseline
2. Percent change of BMI from baseline

Cardiometabolic Risks

1. Percent change of cardiometabolic score (UKPDS risk score for CHD) from baseline
2. Percent changes of cardiometabolic risk factors (SBP, DBP, LDL, total cholesterol, urine microalbuminuria) from baseline

Quality of Life RAND (SF-36)

1. Change from baseline in SF-36 total health score.
2. Change from baseline in SF-36 two main domains, physical and mental health component scores, and in eight distinct subscales (physical functioning, role-physical, bodily pain, general health, vitality, social functioning, role-emotional, and mental health).

Microvascular or macrovascular events

1. Development of stroke, retinopathy, nephropathy, foot ulcer

4.4 Study Outcomes/Adverse Events

We will collect the study outcomes listed in Table 1 below, which are grouped into fifteen clusters. Study outcomes are endpoints related with the initial interventions. Meanwhile we will collect adverse events that are endpoints related with current study protocol (see Section 9.1 for more details). Detailed definitions and algorithms for determining these endpoints are described in the Manual of Operations. Some endpoints are also secondary study endpoints.

These will be self-reported and will be verified with medical records. Each outcome/AE and whether it is verified will be recorded in the CRFs.

We will also collect alcohol usage data using AUDIT-C, the Alcohol Use Disorders Identification Test (Appendix 4). Although this is an observational study, recommended guidelines for clinical action to be considered are outlined in the Manual of Operations, should a resulting AUDIT-C score meet study standardized thresholds for concern.

Table 1. Study Outcomes in ARMMS-T2D

Surgical procedures	New bariatric surgery Abdominal surgery	Endoscopy Orthopedic surgery
Endocrine related	Severe hypoglycemia	Severe hyperglycemia
Cardiovascular	Myocardial Infarction Cardiac re-vascularization	Unstable angina Congestive heart failure
GI tract related	Bowel obstruction Marginal ulcer Stricture Anastomotic leak Intra-abdominal bleeding Gallstones/cholecystitis Internal hernia Ischemic bowel Band intolerance Band erosion Gastric outlet obstruction Colon polyps	Gastric prolapse Dumping syndrome Abdominal wall hernia Abdominal pain GI fistula GERD Hiatal hernia Intussusception Bacterial overgrowth GI bleeding (transfused) Nausea/vomiting
Eye related	Diabetic visual or eye changes Retinopathy	Glaucoma Incident blindness
Nutritional	Iron deficiency Vitamin B12 deficiency Vitamin D deficiency Hyperparathyroidism	Anemia (blood transfusion) Dehydration (IV treatment) Hypercalcemia Hyperkalemia
Renal	Renal insufficiency Renal failure Kidney stones	Macroalbuminuria Dialysis Transplantation
Respiratory	Respiratory failure Pneumonia	Pulmonary edema
Cerebrovascular	Stroke	Transient ischemic attack
Neurological	Depression Neuropathy	Memory loss/cognitive dysfunction
Musculoskeletal	Lower limb fracture Upper limb fracture Other fracture	Spine fracture Joint replacement
Vascular	Diabetes-related foot ulcers or amputation	Blood clots (legs, lungs)
Hepatic or liver related	NASH Cirrhosis	Hepatic insufficiency
Cancer (non-skin)		
Other	Sepsis	Hospitalization in a nursing care facility

4.5 Exploratory Endpoints

Exploratory endpoints include:

1. Biomarkers including C-reactive protein and others
2. Long-term rates of diabetes-related events (e.g., diabetes-related amputation, and for retinopathy: need for laser therapy or use of injected VGEF inhibitors, or incident blindness)
3. Obesity-related events (e.g. new non-skin cancer, etc.)

5.0 SUBJECT SELECTION AND WITHDRAWAL

5.1 Inclusion Criteria

Participants who underwent bariatric surgery or medical/lifestyle therapy in an RCT at one of the 4 participating sites are eligible for this longer-term follow-up study. Participants who enrolled in an RCT at one of the four consortium sites must also have initiated intervention procedures. Participants who were randomized in one of the RCTs but never initiated intervention will not be included.

Current Inclusion Criteria

- Participant must have met original inclusion criteria for study participation and enrolled in an RCT at one of the 4 participating sites.
- Participant must have initiated intervention and / or received randomized treatment after enrolling in the original RCT at one of the 4 participating sites.
- Participants must be willing to comply with the additional follow-up visits and telephone encounters, and sign the informed consent.
  Participants who were initially randomized to medical therapy and later had bariatric surgery are eligible.

5.2 Exclusion Criteria

- Did not receive randomized treatment in one of the four RCTs
- Refusal to sign informed consent

5.3 Participant Recruitment

All participants who have met all protocol inclusion criteria and no exclusion criteria will be eligible for re-consenting. Participants initially randomized to the Medical Management arm who later had bariatric surgery will be eligible for re-consent and will be followed per protocol.

Each site will be responsible for contacting their participants. All eligible participants will be mailed a letter thanking them for their previous participation and asking them to consider continued long-term observation at their original site. The letter is followed in approximately 1 week by a telephone call by study staff to further discuss the longer-term observational follow-up. If participants do not respond to either the initial letter or phone call, sites will send the initial letter via certified registered mail. Participants who agree to enroll will need to provide written consent to the new study procedures according to the conventions of each site's IRB.

Enrolled participants who identify they are unable to attend study visits and therefore are potential for either lost-to-follow-up or study withdrawal will be approached to allow the continued capture of study data through telephone visits and medical record review. Change in Participation Status level will be allowed and the process of study documentation will be completed as specific to this sub-population.

Consistent with procedures used in our ongoing and prior studies, all participants will be consented to allow the study team access to their long-term medical record data, especially as it relates to their pharmacy, laboratory data, and primary care physician, endocrinologists, cardiologists, and eye care notes, so that diabetes, cardiovascular, and microvascular disease status of any participants who are unable to attend study visits or are lost to follow-up at study visits, but do not actively withdraw informed consent, can be tracked through medical-record data and public records. With local IRB approval, and completion of necessary application processes, sites such as National Death Index (NDI) may be utilized to determine whether persons have died and if so, to obtain copies of the matching death records from state vital statistics office, solely for statistical purposes in medical and health research.

5.4 Early Withdrawal of Participants

Participants may withdraw from the study at any time without prejudice, or they may be withdrawn at any time at the discretion of the investigator for safety, behavioral, or administrative reasons. All participants will be followed until the conclusion of the study, unless the subject requests withdrawal of consent either in writing or as documented in study record by study staff.

5.5 Data Collection and Follow Up for Withdrawn Participants

If a participant discontinues or withdraws prior to study completion, all applicable activities scheduled for the annual study visit will be performed at the time of discontinuation, if the participant agrees. Any adverse experiences at the time of discontinuation/withdrawal will be recorded.

6.0 STUDY TREATMENTS

6.1 Description

The ARMMS-T2D observational follow-up study is not an intervention study. We recognize that the medical/lifestyle arms of the interventions differed in each contributing parent trial. Importantly, all of the medical/lifestyle approaches were successful in reducing body weight and improving glycemic control. The study team will monitor diabetes status at study visits and the subject’s primary care provider or primary endocrinologist will be responsible for diabetes management per clinical care guidelines. The study team will notify the subject’s primary care provider (and primary endocrinologist, as needed) regarding their study participation, using a standardized letter focused on diabetes status, any urgently needed changes made to their diabetes therapy at the study visit, or recommendations to chronic therapy. A copy of the study visit report and laboratory assessment will be provided to the study participant and participant’s primary care physician. If the subject does not have a primary care provider, assistance will be provided to identify one.

6.2 Glycemic Target

The HbA_{1c} target for study participants will be <6.5%, unless there is high glycemic variability or recurrent or severe hypoglycemia. This approach is consistent with the American Diabetes Association individualized guidelines.

7.0 STUDY PROCEDURES

In addition to laboratory testing to monitor diabetes control, additional measures will be taken to evaluate quality of life and health status, including end-organ damage and metabolic changes in the medical and surgical groups during the long-term follow-up period.

7.1 Pre-Study Visit

Prior to study entry, and after participants have signed consent for study participation, participants will be contacted (via letter, email, or telephone call) to confirm current contact information and to schedule an annual in-person visit. Sites may review medical records to collect pertinent historical data during the intervening period between completion of the previous study and enrollment into the ARMMS-T2D follow-up study.

Sites can contact participants between date of consent and their first annual visit, as needed, to retain participants. Relevant data collected (i.e. significant adverse events) can be entered on the appropriate unscheduled visit form.

7.2 Standardization of Labs within the Consortium

Some of the sites did not obtain and analyze all of the laboratory tests at baseline and year 1 that are currently listed in this protocol for the annual research visits. When possible, sites should supply laboratory test results for tests that were not included in their individual site protocols from analysis of stored plasma and serum samples, or from the medical record if labs were collected during the intervening period between completion of the original study and enrollment into the ARMMS-T2D follow-up study.

7.3 Annual Research Visit (± 3 months*)

Annual follow-up visits will be scheduled ±3 months of the participant's annual anniversary (i.e., 12-month time point, ±3 months) from entering his or her respective RCT.

Annual follow-up visits should be performed at each site's clinic / office, but if not possible, participants should be followed through other contacts as permitted / consented by the subject (i.e., remote study visits, primary care physician, medical record review, professional laboratory service or in-home specimen collection kits) and as allowed by the individual site IRB or ethics committee.

The following procedures will be performed at each annual visit

- Anthropometric Measures and Vital Status:
    - Height
    - Weight
    - Sitting BP
    - Waist Circumference
- Central Laboratory Tests:
    - Fasting blood glucose
    - Fasting insulin
    - HbA_{1c}
    - CBC (with differential and platelet count)
    - Chem 20
    - Fasting C-peptide
    - Vitamin D
    - PTH

    - Fasting lipid profile (total cholesterol, LDL, HDL, triglycerides)
    - Iron
    - Ferritin
    - TIBC
    - Vitamin B12
    - Markers of inflammation (hsCRP)
    - Urinalysis (micro albumin and creatinine) from morning-void
- Labs for Research Repository / Biobank (if consented)
- Quality of Life Survey (RAND SF-36)
- Medical History
- Medication Usage and Vitamin/Iron Supplement
- Alcohol Use Disorders Identification Test (AUDIT-C)
- Record Study Outcomes or Adverse Events
- Schedule Next Visit
- Thank you letter and visit summary

The preferred window is ± 3 months, but annual visits will be accepted in the extended window of up to ± 6 months. Height will be only measured at the initial annual visit.

7.4 Interim Communications and Retention Activities

Three interim visit communications will be performed between each annual visit to help retain participants in the study and improve communication with them. Additional interim communication may occur to improve retention at the discretion of the study staff and with approval by local IRBs as required. All participants will be contacted at the time points indicated below to review health status, change in medications, hospitalizations, study outcomes, AE's / SAE's and other visit scheduling and retention activity related information

- Interim Visit One (3 months after each annual visit, ±1 month):
    - Mail a reminder to schedule the next year's in-person annual visit, followed by telephone or email contact to schedule that visit.
    - Contact all participants to review health status, change in medications, hospitalizations, study outcomes, AE's / SAE's and thank them for their participation.
- Interim Visit Two (6 months post annual visit, ±1 month):
    - Send a thank-you "mailer" to the participant in recognition of his or her participation.
    - Schedule the next annual visit if not scheduled before
    - The mailer may include an updated patient newsletter
    - Contact all participants to confirm contact information
    - Review health status, change in medications, hospitalizations, study outcomes, AE's / SAE's.
- Interim Visit Three (9 months after the first annual visit, ±1 month):
    - Contact all participants to confirm their contact information and remind them of their next annual visit.
    - Review health status, change in medications, hospitalizations, study outcomes, AE's / SAE's and thank for participation.

The preferred window is ±1 month, but interim visits will be accepted in the extended window of up to ± 1.5 months.

7.5 Schedule of Visits/Procedures

Table 2. Schedule of Activities

Visit	Screen/ Re-consent	Historical Data (Pre-Study)	Each Annual Visit	Interim phone follow-up visits, Post 3, 6, 9 mo. annual visit	Subsequent Annual Visits
Window*			± 3 months	± 1 month	± 3 months
Informed Consent	X				
Physical Measurements					
Height			X**		
Weight, Waist Circumference		X	X		X
Blood Pressure		X	X		X
Medical History					
History of HTN		X			
History of CVD		X			
Medications					
Anti-Diabetic		X	X	X	X
Cardiovascular		X	X	X	X
Anti-hypertensives		X	X	X	X
Prescription Weight Loss		X	X	X	X
Vitamin Supplement		X	X	X	X
Labs					
Fasting glucose		X	X		X
Fasting insulin		X	X		X
HbA_{1c}		X	X		X
CBC w/diff and platelets			X		X
Chem 20+			X		X
Fasting C-peptide			X		X
hs CRP			X		X
Vitamin D			X		X
PTH			X		X
Lipid Profile			X		X
Iron			X		X
Ferritin			X		X
TIBC			X		X
B12			X		X
Urinalysis++			X		X
Research repository+++			X		X
Quality of Life					
SF-36 (RAND 36-Item)			X		X

	Safety Endpoints++++				
Study Outcomes/Adverse Events		X	X	X	X
AUDIT-C		X	X	X	X

* The preferred window is ± 3 (1) months, but an extended window of up to ± 6 (1.5) months is also accepted
** Height will be only measured at the first annual visit
+ Albumin, Alkaline phosphatase, ALT (IU/L), AST (IU/L), BUN, Calcium-serum, Serum chloride, CO2, Creatinine (mg/dL), Direct bilirubin, Gamma-GT, Glucose test, LDH, Phosphorus, Potassium, Serum Sodium, Total bilirubin, Total cholesterol, Total protein, Uric acid
++ Microalbumin:creatinine ratio on morning-void urine
+++ Per subject consent
++++ Defined in Section 4.4

7.6 Study Assessments

7.6.1 Laboratory Procedures

Table 2 shows the schedule of blood samples that will be collected at every annual study visit. A qualified central lab will be used for clinical laboratory assessments. Under circumstances that the blood samples cannot be sent to the Central Lab, data can be collected from local (CAP / CLIA) approved labs. All urinary albumin measures are made on a morning void prior to vigorous exercise. In addition to performing a standard urinalysis, the urine sample will be tested for albumin and creatinine.

Subject to IRB approval and signed subject informed consent, samples for biobanking / research repository should be drawn and frozen at each site for later analysis. Participation in the research repository is optional. Participants may participate in the main study even if they choose not to participate in the biobanking / research repository evaluation. Participants, who no longer wish to participate in the biobanking/research repository or no longer want their blood stored/analyzed for future research, need to specifically withdraw consent from the biobanking / research repository study. Blood, serum and plasma samples will be stored indefinitely or until all samples are used up.

7.6.2 Physical Measurements and Vital Signs

Height will be measured at the first annual visit. Weight, waist circumference and blood pressure will be measured at each annual visit.

For height and weight, participants will be weighed in light clothing, with empty pockets, and without shoes. For waist measure, we will use the NHLBI method (108).

7.6.3 Quality of Life Assessment (SF-36)

The RAND 36-Item Health Survey will be used for the SF-36 quality of life assessment. The SF-36 is a multi-purpose survey of general health and outcomes. It uses patient responses to measure each of eight health concepts, and can be scored as an 8-scale profile of summary physical and mental health outcomes. It also includes a self-evaluation of outcomes, which have been shown to capture changes in health during the past year. Participants will complete the SF-36 at each annual visit (Appendix 3). If participants are not able to complete the annual visit at the research center, a SF-36 form will be mailed to the subject for them to complete.

7.6.4 Study Outcomes and Adverse Events

All study outcomes/adverse events will be collected and reported on the corresponding page of the CRFs from date of consent. Serious study outcomes and adverse events must be reported immediately as described under section 9.1. The Alcohol Use Disorders Identification Test (AUDIT-C) will be used to assess alcohol use disorder.

7.6.5 Public Health Emergencies

Declarations of a public health emergency may be made if a disease or disorder presents a public health emergency; or if a public health emergency, including significant outbreaks of infectious disease such as the COVID-19 pandemic, exists. Revisions may need to be made to action plans and strategies for facilitating participant recruitment, study visits, data and specimen collection and retention due to such emergencies.

The goal of revisions should be to assure the safety of the study participants, maintain compliance with good clinical practice (GCP), and minimize risks to the integrity of the study. Sponsor guidance, as well as federal and local regulations should be followed. Revisions may include, but will not be limited to remote telecommunications and telehealth (virtual) visits between study participants and study.

Handling a crisis of this magnitude must remain fluid. Revisions may need to be changed over time and should therefore, undergo periodic review to maintain continuity and responsiveness to the overall crisis management.

8.0 STATISTICAL PLAN

8.1 Analysis Plan for Aim 1

Aim 1 – To compare the durability of glycemic control, assessed by HbA$_{1c}$, between patients randomized to bariatric/metabolic surgery or comprehensive medical/lifestyle therapy

In general, patient baseline characteristics will be summarized with appropriate descriptive statistics. Continuous characteristics will be compared between surgical and medical groups using linear regression models that include site as a covariate. Skewed continuous variables will be transformed appropriately (e.g., with log transformation). Categorical variables will be compared using the Cochran-Mantel-Haenszel test.

Analysis of primary outcome

The primary outcome analysis will focus on comparing the percent change in HbA$_{1c}$ from baseline between the surgical and non-surgical groups. Since the data are longitudinal and have hierarchical structure, we will use a linear mixed-effects model for the analysis. The fixed effects of the model will include group (surgical *vs*. non-surgical), yearly visit, and an interaction between group and visit. Furthermore, to account for possible heterogeneity of the four parent studies, we will include site as a fixed covariate in the model. The model will include a random intercept at the patient level to account for correlations due to the hierarchical data structure. More specifically, the model will take the following mathematical form,

$$\boldsymbol{Y_{ijt}} = \boldsymbol{\alpha_t} + \boldsymbol{\beta_t X_{ij}} + \boldsymbol{\gamma^T Z} + \boldsymbol{b_i} + \boldsymbol{c_{ij}} + \boldsymbol{e_{ijt}} \quad \text{(A1)}$$

where $\boldsymbol{Y_{ijt}}$ is the percent change of HbA$_{1c}$ for subject j=1, …, n_i from site i=1, …, 4 at visit t; $\boldsymbol{X_{ij}}$ is a binary indicator for treatment group (1 for surgical group and 0 for non-surgical group); $\boldsymbol{Z}$ is a 4-dimensional vector with the ith element equal to one as the subject is from the ith site; $\boldsymbol{\alpha_t}, \boldsymbol{\beta_t}$ and $\boldsymbol{\gamma}$ are unknown coefficients; $\boldsymbol{b_i}$ represents a fixed effect for site i; and $\boldsymbol{c_{ij}} \sim \boldsymbol{N}(\boldsymbol{0}, \boldsymbol{\sigma_c^2})$ is a Gaussian random effect at the patient level; the residual vector, $\boldsymbol{e_{ij\cdot}}$, follows a multivariate-Gaussian distribution $\boldsymbol{MVN}(\mathbf{0}, \boldsymbol{\Sigma_e})$; The form of $\boldsymbol{\Sigma_e}$ will be selected with the Akaike Information Criterion. The regression coefficients should satisfy appropriate constraints for model identifiability. Under this model, $\boldsymbol{\beta_7}$ represents the difference in the mean percent changes of HbA_{1c} at year 7 for patients in the surgical and non-surgical groups. Testing the hypothesis

$$\boldsymbol{H_0}: \boldsymbol{\beta_7} = \mathbf{0}$$

will provide the primary analysis results. The hypothesis will be rejected with a two-sided p-value < 0.05.

Some patients will have been followed for up to 13 years at the end of this proposed study (e.g., STAMPEDE and TRIABETES patients). In addition to year 7, the model (A1) allows for estimating the HbA_{1c} changes of the two groups and their differences at each year, which will be performed as a secondary analysis to illustrate the trend over the entire follow-up period.

Analysis of secondary outcomes

In addition to the primary outcome of HbA_{1c} percent change, several secondary outcomes will also be analyzed, including a) glycemic control, b) diabetes remission and partial remission, and c) diabetes relapse after initial remission. The definitions of the secondary outcomes can be found in Table A1. Since the secondary outcomes of glycemic control and diabetes remission are longitudinal binary data, their analysis will be performed with the generalized linear mixed-effects models (mixed-effects logistic regression models). More specifically, the model for glycemic control will take the following form,

$$\log \frac{\boldsymbol{P_{ijt}}}{\mathbf{1} - \boldsymbol{P_{ijt}}} = \boldsymbol{\alpha_t} + \boldsymbol{\beta_t X_{ij}} + \boldsymbol{\gamma^T Z} + \boldsymbol{b_i} + \boldsymbol{c_{ij}} \quad \text{(A2)}$$

where $\boldsymbol{P_{ijt}}$ is the probability of achieving glycemic control for subject j=1, …, n_i from site i=1, …, 4 at visit t; other notations in model (A2) follow those in model (A1). Under this model, $\boldsymbol{\beta_7}$ represents the log odds ratio of achieving glycemic control at year 7 for patients in the surgical group in comparison with those from the non-surgical group. Testing the hypothesis $\boldsymbol{H_0}: \boldsymbol{\beta_7} = \mathbf{0}$ is the main analysis for the secondary outcomes. Estimates at other years will also be derived.

The analysis of diabetes remission and partial remission will be the same as that for glycemic control. The analysis of diabetes relapse will be only descriptive by estimating the Kaplan-Meier curves. Group comparison will not be performed since the non-surgical group is expected to have a very small number of remissions and thus relapses.

Power analysis for Aim 1

Our combined data show that the percent changes of HbA_{1c} from baseline to 1-year are -24.0% and -9.1% for the surgical and non-surgical groups, respectively; and the percent changes are -21.6% and -0.1% at year 3 based on the combined STAMPEDE and TRIABETES data. STAMPEDE has finished 5 years of follow-up so far, and the data show an approximately -20% change in the surgical group (about -2% in the non-surgical group). For the power analysis, we assume that the mean HbA_{1c} percent change is -20% at year 5 in the surgical group, and it increases linearly to -15% at year 7. We also assume a linear trend for the HbA_{1c} percent change over time in the non-surgical group, and its 7-year percent change ranges from -7% to +2% for different rates of crossovers to the surgical group; to date, the observed crossover rate is 0.8%, but this might increase over time. Since the primary analysis will employ the intention-to-treat (ITT) principle, a larger crossover rate may lead to a lower HbA_{1c} at year 7 for the non-surgical group.

A total of 355 patients were originally randomized across the four sites, 325 initiated treatment, 20 (6%) have been lost to follow-up, and 3 (0.9%) died. With our intensive re-recruitment and long-term retention strategies, we expect to re-enroll and achieve 7-year follow-up on at least 80% of the 302 patients in the available cohort (i.e., n≈242). The statistical power for testing the hypothesis $\boldsymbol{H_0}$ under the linear mixed-effects model (A1) does not have a closed form. Therefore, we used Monte Carlo simulations to estimate the power. The values of the variance components of the mixed model were estimated by fitting the current data (σ_b=0.06, σ_c=0.15, σ_e=0.10 and ρ=0.13, assuming an auto-regressive structure with order one for the matrix $\boldsymbol{\Sigma_e}$). Figure A1 shows the power for the primary hypothesis under different simulation scenarios, where N represents the number of patients with outcome data observed at year 7. Each power is a percentage achieving statistical significance in testing $\boldsymbol{H_0}$ above at a two-sided alpha level of 0.05 among 500 simulations. The power is higher than 90% except for the scenario with 30% missing data.

Power for secondary outcomes

For the secondary outcomes of glycemic control and diabetes remission, the powers were estimated under the mixed-effects logistic models using the simulation method proposed in Dang et al. **(109)**. (i) For *diabetes remission*, the combined 1-year data show the remission rates are 46% and 1% in the surgical and non-surgical groups, respectively. The combined 3-year data from STAMPEDE and TRIABETES show a 32% remission rate at year 3 for the surgical patients (0% for the non-surgical patients). For the power analysis, we assume that the remission rate is 46% at year 1 and its log odds declines linearly, yielding a rate of 20% at year 7 in the surgical group. For the non-surgical group, we assume a 5% rate to account for possible crossovers to the surgical group. Based on 500 simulations, the power was estimated to be 94.6% (two-sided alpha=0.05, 20% missing rate). (ii) For *partial remission*, we assume that the rates are 27% and 7% at year 7 for the surgical and non-surgical groups, respectively, based on the preliminary data. The power for detecting such a difference was 98.6%. (iii) For the rate of *glycemic control*, our preliminary data show 75% and 52% for the surgical and non-surgical groups at year 1, respectively, 63% and 38% at year 3, and 48% and 22% at year 5. Assuming linear curves for the log odds of the glycemic control rates for both groups (consequently, 39% vs. 11% at year 7), the power was estimated to be >99%. The power is still about 98% even with a 40% missing rate. (iv) Power analysis was not performed for diabetes relapse since its analysis will be descriptive.

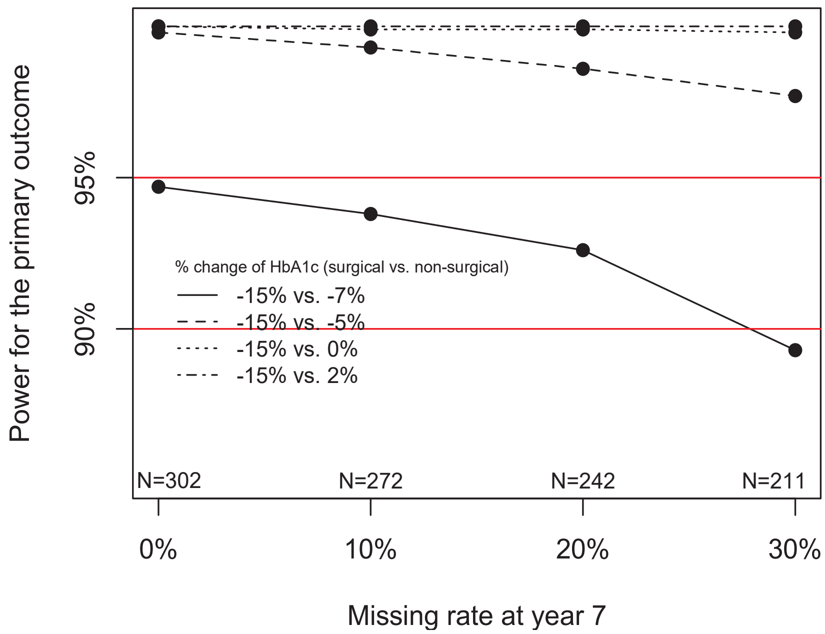


Figure A1. Power for analyzing the primary outcome in Aim 1

The missing data pattern for longitudinal studies is typically complicated, including intermittently missing data, dropouts and other patterns. For simplicity, the simulation discarded participants with any missing data. Therefore, the actual power is expected to be even greater since the analysis will include all the available data.

Table A1. Definitions of study outcomes by aims

Aim		Outcome	Definition	Variable type
Aim 1	Primary	HbA_{1c} change	Percent change from baseline	Continuous
	Secondary	Glycemic control	HbA_{1c} <7.0%	Binary
		Diabetes remission	HbA_{1c} <6.5% without DM meds	Binary
		Partial remission	HbA_{1c} <6.5% with or without DM meds	Binary
		Diabetes relapse[1]	Time from the initial remission to relapse	Time-to-event
Aim 2	Efficacy	Body weight	Percent change from baseline	Continuous
		BMI	Percent change from baseline	Continuous
		Cardiometabolic risk score[2]	Percent change from baseline	Continuous
		Quality of life	Percent change from baseline	Continuous
		Cardiometabolic risk factors (SBP, DBP, LDL, total	Percent change from baseline	Continuous

		cholesterol, urine microalbumin[3])		
		Microvascular or macrovascular events	Stroke, retinopathy, nephropathy, foot ulcer	Binary
	Safety[4]	Safety endpoints	Severe hypoglycemia and hyperglycemia, major adverse cardiovascular events, etc.	Binary
Aim 3		Time to first remission		Time-to-event

[1]Relapse is defined as HbA_{1c} ≥6.5% and/or restarting diabetes medications

[2]UKPDS risk score for CHD (Stevens et al., 2001)

[3]Microalbumin data were not collected at all sites and the analysis will be restricted to patients with data measured.

[4]See Table 1 for the full list

Sensitivity Analyses

Missing data

Both the linear and generalized linear mixed-effects models account for missing data by implicitly assuming data missing at random. Missing data in this study will include intermittently missing pattern and monotone missing pattern caused by dropouts (lost to follow-up or death). A minimal death rate (0.9% to date) is expected for this study. In general, all methods accounting for missing data depend on somewhat untestable assumptions **(110)**. Maximal efforts will be made to minimize missing data, in particular the provisions to obtain the key 7-year outcome data. We will also perform the following sensitivity analyses.

(1) *Multiple imputation*. We will use multiple imputations by chained equation (MICE) to impute the missing endpoints **(111, 112)**. To do so, the entire dataset will be organized as a rectangle with columns representing baseline characteristics and longitudinal endpoints arranged chronologically. Each mixed model will be applied to multiple imputed datasets, and results will be summarized using Rubin's rule.

(2) *Pattern mixture models*. Pattern mixture approach **(113, 114)** is especially useful in the case of informative dropout, where the dropout times (e.g., death time) and subsequent censored endpoints depend on actual missing data. In this model, the probability distribution of the dropouts may further differentiate those caused by death and other mechanisms.

Crossovers

We recognize that some patients might crossover from the non-surgical arm to a bariatric surgery option sometime during the follow-up period. There might also be a very small number of crossovers from the surgical group to a non-surgical arm. The primary analysis will employ the intention to treat (ITT) principle, which considers each patient in the group assigned at randomization, regardless of later crossovers. To account for possible bias introduced by crossovers, we will perform a sensitivity analysis based on the *IPTW (*inverse probability-of-treatment weight) approach. Since crossover is likely to depend on previous outcomes, considering the treatment pathway during the study period as a simple time-dependent variable might be biased. Therefore, we will use the marginal structure models (MSM) with IPTW to account for the treatment crossovers. Stabilized IPTWs **(115)** will be used to weight each observation in the analysis. The weights will be calculated with logistic regression models with previous treatment history and baseline covariates. Participants who do not crossover will have a weight of one. Another advantage of MSM is that it can handle crossover and informative censoring simultaneously. A censoring weight can be computed similar to IPTW. The analysis then weighs each subject with a final weight, which is a product of IPTW and the censoring weight. Finally, a per protocol sensitivity analysis will also be performed by discarding any outcomes observed after crossovers. Given our sample size and the large differences in HbA_{1c} and secondary outcomes that we have observed through 5 years, we expect these sensitivity analyses will have a minimal effect on the analyses.

Exploratory Analyses

In addition to the primary analysis, we will also conduct an exploratory analysis to compare individual surgical and non-surgical interventions by employing direct and indirect comparison methods similar to those in network meta-analyses **(116)**. Direct comparison applies to two interventions from the same study, whereas indirect comparison applies to interventions from different studies. More specifically, we will examine and compare four interventions: Roux-en-Y gastric bypass (RYGB), laparoscopic adjustable gastric banding (LAGB), sleeve gastrectomy (SG), and the non-surgical intervention. For this analysis, we will revise model (A1) by replacing the binary indicator variable for surgical and non-surgical groups with a 4-category variable representing these four individual interventions. The revised model for this exploratory analysis will have the following form:

$$\boldsymbol{Y_{ijt}} = \boldsymbol{\alpha_1^T D_{ij}} + \boldsymbol{\alpha_2^T T_{ij}} + \boldsymbol{\alpha_3^T Vec}\left(\boldsymbol{D_{ij} T_{ij}^T}\right) + \boldsymbol{\gamma^T Z} + \boldsymbol{b_i} + \boldsymbol{c_{ij}} + \boldsymbol{e_{ijt}}, \quad \text{(A3)}$$

where $\boldsymbol{D_{ij}}$ is a 4-dimensional vector with the *l*th element equal to 1 if the patient is in the *l*th intervention group and all other elements equal to 0; $\boldsymbol{T_{ij}}$ is a vector with the *t*th element equal to 1 (i.e., at *t*th year visit) and all other elements equal to 0; $\boldsymbol{Vec}(\boldsymbol{D_{ij} T_{ij}^T})$ is a vector of the matrix $\boldsymbol{D_{ij} T_{ij}^T}$ to represent all interactions between year and interventions. Other notations are the same as those in model (A1). For each year, the least-square estimates of HbA_{1c} percent changes of the four groups will be obtained by fitting model (A3). We will use the Bonferroni procedure to adjust for multiple pairwise comparisons between individual interventions.

The linear mixed-effects models used for analyzing the primary outcomes account for study heterogeneity by assuming that the percent changes of HbA_{1c} in the surgical/non-surgical groups at the four sites are not identical, but follow some distribution. The final estimate of the effect (difference of the percent changes of the two groups) from the mixed-effects models represents an average of the effects at the four sites, and the confidence interval associated with the estimate describes the degree of heterogeneity **(117)**.

In addition to the classic mixed-effects model approach, we will perform an additional exploratory analysis using a Taxonomy approach. This novel methodological approach is used to address study heterogeneity by incorporating intervention components that are grounded in common theory across sites, rather than the original interventions **(118-120)**. We have employed task analysis and a group consensus approach to successfully decompose the non-surgical interventions at the four sites into four components: exercise, diet counseling, medication adjustment, and other behavioral/lifestyle modifications. Task analysis represents a formalized process for decomposing and describing an activity **(121)**. The decomposed intervention components are summarized in Table A2 below (refer to individual study protocols for more details). Overall, as Table A2 shows, the four studies have very similar approaches – each employed an intensive program of exercise, diet, behavioral, and medical intervention. For these four components, the doses (e.g., exercise time) varied somewhat across the sites, but in each case the dose delivered is far above what would be expected in usual clinical care settings for patients with diabetes. For descriptive purpose, the doses in Table A2 are the average intervention times that each patient is expected to spend according to the corresponding protocol.

Table A2. Average doses[1] of the decomposed intervention components by site

		Site			
		CROSSROADS	SLIMM-T2D	STAMPEDE	TRIABETES
Exercise	Supervised exercise	176 hrs/year	12 hrs/year	0 hrs/year	0 hrs/year
	Unsupervised exercise	196 hrs/year	222 hrs/year	183 hrs/year	174 hrs/year
	Total exercise	372 hrs/year	234 hrs/year	183 hrs/year	174 hrs/year
Diet	Nutrition Counseling	32 hrs/year	22 hrs/year	2 hrs/year	28 hrs/year
Medication	Medication use[2]	Available	Available	Available	Available
	Medication dosages[2]	Available but requires source document review	Available but requires source document review	Available but requires chart review	Available but requires source document review
	Care provider	PCP	Study investigator	Study investigator	PCP
Other lifestyle modification		26 hrs/year	22 hrs/year	2 hrs/year	28 hrs/year

[1]For exercise, diet, and other lifestyle modification components, the doses were the average time (in hours) that each patient should have spent on these interventions within the first one-year period according to study protocols. Obtaining the actual doses (times) for each patient will require reviewing source documents such as attendance logs.

[2]The current ARMMS-T2D data registry collects information on medications for diabetes including insulin, biguanides, thiazolidinediones, incretin mimetics, secretagogues, alpha-glucosidases, DPP-4 inhibitors and other approved drugs, as well as non-diabetic medications including beta blockers, statins/lipid modifiers, ACE Inhibitors, ARBs, anticoagulants, diuretics, digoxin, nitrates, calcium channel blockers, other anti-hypertensive and prescription weight-loss medicines.

Among the 4 sites, the surgical procedures (n=192) include RYGB (n=104), SG (n=49), and LAGB (n=39), (Table A3). RYGB was performed with slight variation at the 4 sites (Table A4). Differences in construction of the RYGB procedure from site to site were small and consistent with standard practice in the US. The pouch size was uniformly 15-20 ml except at UW (40 ml). The Roux-limb length was 150 cm at all sites except Brigham and Women's (75 cm). The biliopancreatic limb length was approximately 50 cm at all sites. The LAGB procedure was performed in identical fashion at the Brigham and Women's and UPMC sites. The SG procedure was performed uniformly at Cleveland Clinic using a 32 French bougie with estimated volume reduction of 75-80%. The slight procedural variation from site to site (RYGB only) is not expected to significantly affect glycemic or weight loss outcome. The surgical procedures are largely considered to be the same. This is supported by observations from Courcoulas and colleagues in the Longitudinal Assessment of Bariatric Surgery (LABS), where it was noted that among 120 pre- or intra- operative factors examined, few pre- and no intra-operative factors except band circumference made any difference in terms of weight loss at 3 years. All the bands in TRIABETES and SLIMM-T2D were Allergan 10 cm or Allergan AP Standard, so none were large circumference bands **(122)**.

Table A3. Number of surgery procedures completed at each site

	STAMPEDE	TRIABETES	SLIMM-T2D	CROSSROADS
RYGB	50	20	19	15
SG	49	0	0	0
AGB	0	21	18	0
Total	99	41	37	15

Total N= 192 for all surgical procedures.

Table A4. Key descriptive characteristics of the RYGB surgery (N=104) at each site

	STAMPEDE (n=50)	TRIABETES (n= 20)	SLIMM-T2D (n=19)	CROSSROADS (n= 15)
Pouch	15-20 ml	20 ml	15-20 ml	40 ml
Roux-Limb	150 cm	150 cm	75 cm	100-150 cm
BP Limb	50 cm	50 cm	50 cm	30-50 cm

The taxonomy data analysis will also be based on linear mixed-effects models as the primary analysis. Basically, the binary indicator (surgical or non-surgical) in model (A1) will be replaced by a set of decomposed intervention measures. In this case, the model has the form as:

$$\boldsymbol{Y_{ijt}} = \boldsymbol{\alpha_t} + \boldsymbol{\beta_t^T W_{ijt}} + \boldsymbol{\gamma^T Z} + \boldsymbol{b_i} + \boldsymbol{c_{ij}} + \boldsymbol{e_{ijt}}, \text{ (A4)}$$

where $\boldsymbol{W_{ijt}}$ is a vector of variables representing the decomposed intervention measures, including surgery type (three dummy variables for RYGB, LAGB and SG, respectively), supervised exercise time (hrs), unsupervised exercise time (hrs), time allocated to nutrition counseling (hrs), number of medications, whether the patient met the study investigator or his own care provider for medication adjustment, the amount of time the patient spent with other lifestyle modification activities (hrs) during the *t*th year. Other notations are the same as those in model (A1). Note that the first three components of $\boldsymbol{W_{ijt}}$, the surgery type, do not vary with *t*. Patients in the surgical groups also received some medical interventions in later follow-up years. There are no *a priori* reasons to hypothesize that a particular intervention component is stronger than others. However, various exploratory analyses could be performed by applying appropriate linear contrasts. For example, $\boldsymbol{\beta}_{71} - \mathbf{100}\boldsymbol{\beta}_{74}$ represents the difference of HbA_{1c} percent changes at year 7 for patients who underwent RYGB alone in comparison to non-surgical patients who exercised 100 hours under supervision per year, where $\boldsymbol{\beta}_{7j}$ is the *j*th component of the parameter vector $\boldsymbol{\beta}_{7}$.

To perform the taxonomy analysis, it is important to obtain data for the intervention measures, i.e., $\boldsymbol{W}_{ijt}$. The current registry database has already collected data for some of these variables, e.g., medication use per year. The data for other variables are available either in the source documents (e.g., attendance logs) or in the electronic medical health records (e.g., billing records). For the ARMMS-T2D U01 follow-up period, the study coordinator at each site will review the source documents and enter the relevant data into the database for all previous years. For future data collection about these intervention measures, form A1 (see Attachment below) will be filled quarterly and data will be then normalized on a yearly basis. In the case that there are more than minimal missing data for some of these covariates (intervention components), which we recognize is possible for previous years of the study; multiple imputation will be used to impute the missing values. As another alternative, we will run a simpler model without using the time-varying covariates $\boldsymbol{W}_{ijt}$ but using summary statistics (e.g., average exercise time per year) of each component as time-fixed covariates.

The exploratory analyses will be performed for the secondary outcomes (glycemic control, diabetes remission and partial remission) as well. In summary, the binary indicator for surgical and non-surgical groups (X_{ij}) in the generalized linear mixed-effects model (A2) will be replaced by D_{ij} (indicator for individual interventions) in model (A3) or W_{ijt} (taxonomy components) in model (A4), respectively.

For all models above, further sensitivity analyses will be performed by including baseline covariates such as HbA_{1c}, BMI, and all other patient baseline characteristics that differ between the two groups ($p<0.05$). An important feature of our study is that a significant portion of the participants have baseline BMI lower than 35 kg/m^2. All analyses above will be performed for this subgroup.

8.2 Analysis Plan for Aims 2 and 3

Aim 2 – To assess the longer-term efficacy and safety outcomes in patients randomized to bariatric/metabolic surgery or comprehensive medical/lifestyle therapy

We will analyze multiple secondary efficacy outcomes in Aim 2, whose definitions are provided in Table A1. Since most secondary efficacy outcomes (e.g., the percent changes of body weight, BMI, cardiometabolic risk and quality-of-life scores, etc.) are continuous variables measured longitudinally, their analyses will be the same as that for the primary outcome (percent change of HbA_{1c}).

The outcome of microvascular or macrovascular events will be summarized as 2-by-2 tables and will be compared using the Cochran-Mantel-Haenszel test. The safety outcomes in Aim 2, including major adverse events and complications, will be analyzed in the same manner.

Power analysis for Aim 2

Table A5 summarizes the minimally detectable effect sizes for the secondary outcomes, under the general assumption that 20% of the patients have missing outcomes.

(1) For longitudinal continuous outcomes, the effect size is defined as the absolute difference of the two group means at year 7 divided by a common standard deviation, assuming an AR(1) structure for $\boldsymbol{\Sigma}_{e}$. The study has a power of 80% to detect an effect size in the range of [0.43, 0.47] with an autocorrelation coefficient in [0.1, 0.8]. For example,

fitting the current body weight data (% of weight change from baseline) from the STAMPEDE study shows that the standard deviation of the residual is 6% and the auto correlation is about 0.7, which translates to a detectable difference of approximately 2.8% between the two groups at year 7. Meanwhile the observed difference is about 20% at year 3 **(62)**. Therefore, we expect to have ample power to analyze that outcome.

(2) For binary outcomes, the minimum detectable odds ratio is 3.03 at 80% power, assuming a 10% event rate in the surgical group.

More scenarios of the power analysis can be found in Table A5. We recognize that the study may not have adequate power for analyzing all these secondary outcomes, especially for some safety outcomes with low event rates. Since these are secondary analyses, the power analysis did not adjust for multiple comparisons for the analysis of multiple secondary endpoints.

Table A5. Minimal detectable effect size for secondary outcomes in Aim 2 (at *power=80%, two-sided type I error=0.05, assuming a 20% missing rate of the outcomes*)

Outcome		Detectable effect size	
Efficacy outcome	Longitudinal continuous variables	Δ=0.47	rho=0.1[1]
		0.43	rho=0.8
Safety outcome	Binary variables	OR=3.03	p=10%[2]
		2.46	p=20%

[1]For longitudinal continuous outcomes, the effect size (Δ) is the absolute difference of the two group means at year 7 divided by a common standard deviation (the diagonal element of the matrix $\boldsymbol{\Sigma_e}$); rho is the autoregressive correlation coefficient.

[2]For binary outcomes, the effect size is the odds ratio (non-surgical vs. surgical) and *p* is the event rate in the surgical group.

Exploratory analyses

As in Aim 1, we will perform similar sensitivity and exploratory analyses for these secondary outcomes in Aim 2, which include analyses of missing data; comparing individual interventions; and adjusting the results for patient characteristics such as HbA_{1c}, BMI, and characteristics that differ between the two groups ($p<0.05$) at baseline.

Aim 3 – To identify clinical predictors of diabetes remission and relapse

In addition to knowing overall rates of remission and relapse, understanding which pre- and early post-operative clinical factors are related to these outcomes will give health care providers key information to identify patients who are most likely to benefit from surgery. Our team has extensive experience with investigations of predictors for a variety of clinical outcomes, including successfully identifying predictors of diabetes remission in smaller samples than our current proposed study **(14, 54, 62, 123-125)**. Toward this goal, we will fit a series of time-to-event regression models with the outcome being the time to first remission. These analyses will be viewed as exploratory, with the goal of identifying the factors that are most important from the broad range of demographic, diagnosis, and treatment data we have access to (e.g., age, gender, race/ethnicity, bariatric procedure type, measures of T2DM severity and duration, as well as measures of hypertension, dyslipidemia, and a composite metabolic syndrome score). Although a large number of variables will be considered, specific variables that are of greatest interest based on the prior published literature include age, diabetes duration, weight loss, insulin use prior to surgery, baseline HbA_{1c}, glucose and insulin levels, procedure type, and the amount of initial weight loss and reduction in BMI **(54, 62)**. First, each variable will be related to the outcome in a univariate model. For multivariate analysis, although numerous variable selection strategies are available, we will use regularization techniques, i.e., adaptive lasso **(126, 127)**. Briefly, these methods start with a “full” model that includes all potential risk factors and estimates their impact on the outcome, while simultaneously satisfying a penalty term that in some cases restricts the magnitude of the effect and in other cases sets it to zero. A key advantage of these methods is that they satisfy the so-called oracle property, which states that the correct model is always chosen asymptotically. We will use the Cox regression framework with covariate effects on the multiplicative scale (i.e. hazard ratios). The proportional hazard assumption will be tested with Schoenfeld residuals. We will also perform sensitivity analyses based on accelerated failure time models that model the log time-to-event with covariate effects on the additive scale. The models will include site as a strata variable and will control for randomized groups. As a secondary analysis for Aim 3, we will also examine the outcome of time to relapse **(14)**. The models will be same as those used for the analysis of time-to-remission, but will be restricted to patients who experienced remission. A further exploratory analysis will use a multi-state regression model to examine predictors for transition probabilities between three diabetes states: initial state, remission, and death. The model will assume a Markovian process and the transition intensities will be modeled with Cox models **(128)**.

The statistical analyses of all three aims will be performed with SAS 9.3 (Cary, NC) and R-studio (Boston, MA).

Power analysis for Aim 3

Table A6 below summarizes the power analysis for Aim 3. Based on our preliminary data, the overall remission rate (i.e., the proportion of all patients, both surgical and non-surgical having first remission) is about 23% so far. Assuming that the overall remission rate is eventually 30%, the study has a power of 80% to detect a hazard ratio as small as 1.36 per one standard deviation (SD) of a predictor of interest (two-sided alpha = 0.05, $R^2 = 0.1$).

Table A6. Minimum detectable hazard ratio per 1SD of the predictor of interest for Aim 3

Overall remission rate	R^2 of the predictor of interest given other covariates*		
	R^2=0.1	R^2=0.2	R^2=0.5
20%	1.46	1.50	1.67
30%	1.36	1.39	1.52
40%	1.31	1.33	1.43

*R^2 is the proportion of variance explained by the multiple regression of the predictor of interest on the remaining covariates. The calculations were performed with PASS 11.

8.3 Subject Population(s) for Analysis

The analysis population will consist of all participants who enroll (consent) to participate in this observational follow-up study. Participants will be analyzed according to their original randomized treatment assignment to follow the intention-to-treat principle.

9.0 STUDY OUTCOMES AND ADVERSE EVENTS

Any instances of study outcomes or adverse events will be reported to the local IRB using the standard forms and procedures that have been established by the IRB.

9.1 Definitions

Study Outcome

A *study outcome* is any new untoward medical occurrence or worsening of a pre-existing medical condition in a participant, which could be related but does not necessarily have a causal relationship with either surgery or medical management of diabetes. An outcome can be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the use of surgery or a medicinal product, whether or not related. The investigators have identified a listing of outcomes (see Table 1) from the literature and clinical experience. Detailed definitions are provided in the Manuals of Operation.

Serious Study Outcome

A study outcome is considered serious if it is fatal or life-threatening; requires or prolongs hospitalization; or produces a disability. More specifically, it can include any of the following:

- death
- a life-threatening experience
- inpatient hospitalization or prolongation of existing hospitalization
- a persistent or significant disability or incapacity
- Important medical events that may not result in death, be life-threatening, or require hospitalization may be considered a serious adverse experience when, based upon appropriate medical judgment, they may jeopardize the participant and may require medical or surgical intervention to prevent one of the outcomes listed in this definition.

All outcomes that do not meet any of the criteria for being "serious" should be regarded as *non-serious study outcomes.*

Severity of Study Outcome

A study outcome is considered to be of moderate or greater severity if it requires medical evaluation (such as additional laboratory testing) or medical treatment; or if it is a serious study outcome.

Unexpected Study Outcome

A study outcome is considered to be unexpected if it is not identified in nature, severity, or frequency of the surgical or non-surgical managements of diabetes for the patient cohort.

Relatedness of Study Outcome

A study outcome is considered to be related to the surgery or medical management if there is a reasonable possibility that the reaction may have been caused by the initial interventions (i.e., a causal relationship between the reaction and the research procedures cannot be ruled out by the investigator(s)). An outcome can be considered to be definitely, probably, possibly, or unrelated, or relatedness may be indeterminate.

Adverse Event

An *adverse event (AE)* is an adverse reaction that could be related to the current study protocol, including blood draw, form completion and any related procedures.

Serious Adverse Event

An adverse event is considered serious if it meets the same criteria for serious study outcomes.

Severity of Adverse Event

An adverse event is considered to be of moderate or greater severity if it requires medical evaluation (such as additional laboratory testing) or medical treatment; or if it is a serious adverse event.

Unexpected Adverse Event

An adverse event is considered to be unexpected if it is not identified in nature, severity, or frequency in the current IRB-approved study protocol or informed consent process.

Relatedness of Adverse Event

An adverse event is considered to be related if there is a reasonable possibility that the reaction may have been caused by the current study protocol.

9.2 Recording and Verification of Study Outcomes/Adverse Events

At each evaluation, participants will be interviewed to elicit potential study outcomes or adverse events. The occurrence of a study outcome or adverse event will be based on changes in the subject's physical examination, laboratory results, and/or signs and symptoms. All study outcomes or adverse events (except Grade 1 or 2 laboratory abnormalities that do not require an intervention), regardless of causal relationship, will be recorded in the case report forms and source documentation.

Medical records should be used to verify each study outcome or adverse event. In the cases that medical records are not available, the events should also be reported and recorded as unverified.

Any serious study outcome or SAE that occurs after the study period and is considered to be possibly related to the study procedures should be recorded and reported immediately to the Principal Investigator at the site.

9.3 Reporting of Serious Study Outcomes / SAE

Investigators will report serious study outcomes or SAEs that are a result of a research-specific procedure or intervention (i.e., venipuncture) according to their IRB reporting policy. All deaths, regardless of relationship to the study, are deemed serious and must be reported as a serious adverse event. The DCC should be notified of serious study outcomes or SAEs within 24 hours of investigator site's discovery. Any instances of adverse events that necessitate reporting as defined above will be reported immediately to the local IRB using the standard forms and/or procedures that have been established by that IRB. The DCC will inform the National Institutes of Health of any serious study outcomes or SAEs.

Adverse reactions that are determined by the investigator to be unrelated to research-specific procedures need not be reported to the local IRB, unless the local IRB requires otherwise.

9.4 Adjudication of Study Outcomes/Adverse Events

An Adjudication Committee that comprises five members from outside the ARMMS Consortium has been established. All members are leading experts in relevant fields – surgery, cardiovascular, endocrine medicine, and nutrition/GI.

The committee will adjudicate major adverse cardiovascular events, severe hypoglycemia and hyperglycemia, and "questionable" serious outcomes or SAEs, including events resulting in death. A questionable event does not match the exact event definition.

For an event to be adjudicated, the sites should send data and related documents to the DCC after removing patient identification information. The DCC will then forward them to the Adjudication Committee. The DCC will also forward the adjudication results to the site for reporting and database entry.

10.0 OBSERVATIONAL STUDY MONITORING BOARD

An Observational Study Monitoring Board (OSMB) will be appointed by NIH/NIDDK to oversee the study. Their main tasks are to ensure that the study is implemented appropriately, monitor patient safety, and that the confidentiality of research data is maintained. The OSMB Charter document will guide the conduct of the OSMB.

The OSMB will be charged with reviewing study data integrity and will provide recommendations for study continuation or cessation. Since this study is for long-term follow-up of interventional RCTs but is no longer an interventional trial *per se*, related adverse events are not anticipated; thus, investigators have elected not to develop formal stopping rules. Instead, the OSMB can elect to stop the study at any point if they feel it is in the best interest of the patients (ethical call). The OSMB will meet either in person or by phone every 6 months as required by NIDDK. The DCC will schedule the OSMB meetings.

11.0 DATA HANDLING AND RECORD KEEPING

11.1 Confidentiality and Privacy

Information about study participants will be kept confidential and managed according to HIPAA requirements. Those regulations require a signed subject authorization informing the subject of the following:

- What protected health information (PHI) will be collected from participants in this study.
- Who will have access to that information and why.
- Who will use or disclose that information.
- The rights of a research subject to revoke authorization for use of PHI.

In the event that a participant revokes authorization to collect or use PHI, the investigator, by regulation, retains the ability to use all information collected prior to the revocation of subject authorization. For participants who have revoked authorization to collect or use PHI, attempts should be made to obtain permission to collect at least vital status (i.e., whether or not the subject is alive) at the end of their scheduled study period.

11.2 Source Documents

Source data is all information, original records of clinical findings, observations, or other activities necessary for the reconstruction and evaluation of the study. Source data are contained in source documents.

11.3 Case Report Forms

As used in this protocol, the term case report form (CRF) should be understood to refer to either a paper form or an electronic data record or both, depending on the data collection method used in this study.

A CRF is required and should be completed for each consented participant. The completed original CRFs are the sole property of the ARMMS-T2D Consortium and should not be made available in any form to third parties, except for appropriate regulatory authorities.

It is the investigator's responsibility to review, ensure completion and approve all CRFs. CRFs must be signed by the investigator or by an authorized staff member. These signatures serve to attest that the information contained on the CRFs is true. At all times, the investigator has final personal responsibility for the accuracy and authenticity of all clinical and laboratory data entered on the CRFs. Subject source documents are the physician's subject records maintained at the study site. In most cases, the source documents will be the hospital's or the physician's chart. In cases where the source documents are the hospital or the physician's chart, the information collected on the CRFs must match those charts.

11.4 REDCap

REDCap will be used as the electronic data capture system for this study. REDCap is a secure, web application designed to support data capture for research studies, providing user-friendly web-based case report forms, real-time data entry validation (e.g. for data types and range checks), audit trails and a de-identified data export mechanism to common statistical packages (SPSS, SAS, Stata, R/S-Plus). The system was developed by a multi-institutional consortium, which includes Cleveland Clinic and was initiated at Vanderbilt University. The database is hosted at the Cleveland Clinic Datacenter. The system is protected behind a login and Secure Sockets Layer (SSL) encryption. There is an audit trail tracking all logins and activities in the database.

11.5 Data Quality and Management

To ensure quality and completeness of the data, REDCap incorporates a data edit check process for Mandatory Check and Range Check fields along with Branching Logic into the programming of the database. Study personnel at the DCC will monitor the data within the system to assure that identified data discrepancies are resolved per the study specific Data Management Plan.

11.6 Previous RCT Data

The baseline data and all prior follow up data from the RCTs will be collected from each site. All previous labs were analyzed in CLIA certified labs. All previous study Protocols will be referenced in the study specific Data Management Plan.

11.7 Records Retention

The Investigators must maintain all confidential study documentation and take measures to prevent accidental or premature destruction of these documents. They will retain the study documents at least three years after the completion or discontinuation of this study. However, applicable regulatory and institutional requirements will be taken into account in the event that a longer period is required.

12.0 STUDY MONITORING, AUDITING AND INSPECTING

The DCC is responsible for monitoring the conduct of this study. The study will be monitored according to the Monitoring Plan. The investigator will allocate adequate time for such monitoring activities and will ensure that the monitor or other compliance or quality assurance reviewer is given access to all study-related documents and study related facilities (for example, pharmacy, diagnostic laboratory, etc.), and has adequate space to conduct the monitoring visit.

13.0 ETHICAL CONSIDERATIONS

The study protocol, consent forms, data collection forms, and recruitment materials, if applicable, will be submitted to each site's IRB. A site may not initiate any patient contact until it has IRB approval. All study personnel will have completed training in the Protection of Human Subjects according to Institutional guidelines.

13.1 Institutional Review Board (IRB)

It is the investigator's responsibility to ensure that the study protocol and informed consent documents are reviewed and approved by the appropriate IRB. Each clinical site will obtain a letter of approval from the IRB before approaching participants. Sites will provide the DCC with copies of the initial IRB approval notice prior to enrolling the first patient, and subsequent renewals, as well as copies of the IRB approved consent and other IRB approved forms.

If, during the study, it is necessary to amend either the protocol or informed consent document, the investigator will be responsible for ensuring that the IRB reviews and approves the amended documents. IRB approval of any procedures must be obtained before implementing new processes or procedures.

13.2 Informed Consent Document and Process

All participants for this study will be provided a consent form describing this study and providing sufficient information for participants to make an informed decision. The informed consent document will inform patients of their right to refuse any release of their protected health information. Each clinical site, according to local IRB requirements, is allowed to modify this informed consent document and make any necessary editorial changes, as long as neither the meaning nor intent of any section is changed.

The investigator or his/her designee (i.e., research coordinator or study nurse) will inform the patient of all aspects of the study pertaining to the patient's participation in it. The process for obtaining informed consent will be in accordance with all applicable regulatory requirements. The informed consent form (ICF) must be signed and dated by the investigator or his/her designee and the patient BEFORE the patient can participate in the study. The subject will be asked to read and sign the IRB-approved ICF. The participant will receive a copy of all signed and dated consent documents, and the originals will be retained in the patient's study file or medical record.

13.3 Subject Information and Consent

Each clinical site is responsible for the confidentiality of the data associated with participants enrolled in this study, in the same manner that it is responsible for the confidentiality of any patient information within its sphere of responsibility. All forms used for the study data will be identified by coded identification number, which will be generated at the clinical center, to maintain subject confidentiality. All records will be kept in locked file cabinets at the clinical centers with access limited to study staff, and all study staff will identify participants via their unique identifier. Clinical information will not be released without written permission of the participant, except as necessary for monitoring by the IRB or OSMB. The participant grants permission to share research data with these entities in the consent document. Federal regulations govern the protection of patient's rights relative to data confidentiality and use of research data.

Consent procedures and forms, and the communication, transmission and stoppage of patient data will comply with individual-site IRB requirements for compliance with The Health Insurance Portability and Accountability Act (HIPAA). The Privacy Rule of HIPAA governs the protection of an individual's identifiable health information. The DCC will ensure that clinical centers associated with the project comply with HIPAA regulations by requiring documentation from the IRBs with the appropriate authorization or consent form. The DCC will maintain copies of all relevant documents from each clinical center. If IRB approvals are not current, data will not be accepted by the DCC. The REDCap data management system will be used to ensure the confidentiality of electronic protected health information. Biospecimens will be labeled with each participant's coded identification number; there will be no protected health information indicated on the label.

14.0 DATA SHARING PLAN

In order to meet the Consortium objectives, members have agreed to share limited data sets and documents from their original randomized trials. All data will be sent to the DCC and stored on secure, access-limited servers. The shared data and documents will be kept confidential among the members and not shared or published externally without the members' consent.

The DCC will deposit the final de-identified study data into the NIDDK central data repository. The Consortium will follow the NIH data sharing policy (https://grants.nih.gov/grants/policy/data_sharing/). Data will be de-identified appropriately according to HIPAA regulations or NIDDK specifications.

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ATTACHMENTS

Appendix 1

Collaborating Sites and Investigators

Name	Role	Institution
John Kirwan, PhD*	Principal Investigator	Pennington Biomedical Research Center
Philip Schauer, MD*	Co-I	Pennington Biomedical Research Center
Sangeeta Kashyap, MD	Co-Principal Investigator, Site Principal Investigator	Cleveland Clinic Foundation
Mary Elizabeth Patti, MD*	Site Principal Investigator	Joslin Diabetes Center
Ashley Vernon, MD*	Co-I	Brigham & Women's Hospital
Donald Simonson, MD⁺	Co-I	Brigham & Women's Hospital
Anita Courcoulas, MD*	Site Principal Investigator	University of Pittsburgh
William Gourash CRNP*	Co-I	University of Pittsburgh
John Jakicic, PhD	Co-I	University of Pittsburgh
David Cummings, MD*	Site Principal Investigator	University of Washington
David Arterburn, MD*	Co-I	University of Washington
Bo Hu, PhD*	DCC Principal Investigator	Data Coordinating Center (Cleveland Clinic)
Gerald Beck, PhD	Co-I	Data Coordinating Center
Janine Bauman	Project Manager	Data Coordinating Center
Karen Teff, PhD*	NIDDK Project Scientist	NIDDK

All are steering committee (SC) members.
*Voting members; ⁺Dr. Simonson leads the taxonomy core in DCC.

Appendix 2

Summary of the Collaborating Sites, Demographics and Available Participants for ARMMS-T2D

Initial Trials: Enrolled Participants					
	CCF	**UPMC**	**Joslin/BWH**	**Joslin/BWH**	**UW**
Trial name	**STAMPEDE***	**TRIABETES**	**SLIMM-T2D (RYGB vs IMWM)**	**SLIMM-T2D (LAGB vs IMWM)**	**CROSSROADS**
# Randomized	155	69	43	45	43
FPFV date	09Mar2007	10Feb2010	10Mar2010	10Mar2010	07Sep2011
LPFV date	25Jan2011	26May2011	20OCT2011	20AUG2013	11May2012
Number underwent intervention†	**154**	**61**	**38**	**40**	**32**
Died	2	0	0	1	0
Lost to Follow-up‡	12	5	3	0	0
Eligible for ARMMS-T2D	**140**	**56**	**35**	**39**	**32**
Crossovers from medical to surgical	2	1	2	0	0
Crossover from LAGB to RYGB	NA	0	0	0	0
Surgical failure	0	0	0	1	0
Baseline Characteristics					
Treatment Groups	RYGB LSG Medical/Lifestyle	RYGB LAGB Medical/Lifestyle	RYGB Medical/Lifestyle	LAGB Medical/Lifestyle	RYGB Medical/Lifestyle
Age (years)	49±8	47±6	51±6	50±10	53±7
Race: White	110 (73%)	54 (78%)	25 (66%)	27 (68%)	23 (72%)
Gender: Female	99 (66%)	56 (81%)	23 (60%)	18 (45%)	22 (69%)
BMI (kg/m^2)	36.7±3.6	35.0±3.1	36.3±3.4	36.3±3.6	37.7±3.6
Duration of DM (yrs)	8.5±5	6.4±5	10.4±6.3	9.3±4.9	6.8±5
Insulin (N, %)	72 (48%)	26 (38%)	23 (60%)	16 (40%)	17 (53%)
HbA_{1c} (%)	9.2±1.5	7.9±1.7	8.5±1.3	8.2±1.2	7.5±0.9
Fasting Glucose (mg/dL)	179±66	171±64	147±53	161±64	150±47
Outcomes at 1 year					

Initial Trials: Enrolled Participants					
	CCF	**UPMC**	**Joslin/BWH**	**Joslin/BWH**	**UW**
Weight loss at 1 Year (%)	RYGB, -27.6% LSG, -25.0% Med/Life, -5.2%	RYGB, -29.1% LAGB, -18.5% Med/Life, -7.6%	RYGB, -26.9% Med/Life, -6.8%	LAGB, -12.7% Med/Life, -8.3%	RYGB, -25.8% Med/Life, -6.4%
HbA_{1c} <6.0% with no DM medication	RYGB, 42% LSG, 27% Med/Life, 0%	RYGB, 40% LAGB, 23.8% Med/Life, 0%	RYGB, 32% Med/Life, 0%	LABG, 6% Med/Life, 0%	RYGB, 60% Med/Life, 5.9%
HbA_{1c} <6.5% with no DM medication	RYGB, 65% LSG, 37% Med/Life, 0%	RYGB, 60% LAGB, 29% Med/Life, 0%	RYGB, 58% Med/Life, 0%	LABG, 6% Med/Life, 0%	RYGB, 60% Med/Life, 5.9%

* Combined STAMPEDE data. † Patient initiated treatment in either the medical/lifestyle group, or underwent surgical intervention. A total of 38 patients dropped from the study at randomization and were not followed. These patients do not qualify for enrollment into ARMMS-T2D. ‡ A total of 20 patients are currently considered lost-to-follow-up across the 4 sites. Significant efforts are being made to re-engage these patients, and UPMC have obtained a verbal commitment from 3 of the 5 at their site. FPFV, first patient first visit; LPFV, last patient first visit. Med/Life, Medical/Lifestyle. DM, diabetes medication.

Appendix 3

SF-36

SF-36 ™ Health Survey

™ This survey was developed at RAND as part of the Medical Outcomes Study.

INSTRUCTIONS: This survey asks for your views about your health. This information will help keep track of how you feel and how well you are able to do your usual activities. Answer every question by marking the appropriate box. If you are unsure about how to answer a question, please give the best answer you can.

1. In general, would you say your health is: (mark one box)

☐ Excellent

☐ Very good

☐ Good

☐ Fair

☐ Poor

2. Compared to one year ago, how would you rate your health in general now? (mark one box)

☐ Much better now than one year ago

☐ Somewhat better now than one year ago

☐ About the same

☐ Somewhat worse now than one year ago

☐ Much worse now than one year ago

The following items are about activities you might do during a typical day. Does your health now limit you in these activities? If so, how much?

(Mark One Box on Each Line)	Yes, Limited a Lot	Yes, Limited a Little	No, Not limited at All
3. Vigorous activities, such as running, lifting heavy objects, participating in strenuous sports	☐	☐	☐
4. Moderate activities, such as moving a table, pushing a vacuum cleaner, bowling, or playing golf. .	☐	☐	☐
5. Lifting or carrying groceries	☐	☐	☐
6. Climbing several flights of stairs.	☐	☐	☐
7. Climbing one flight of stairs	☐	☐	☐
8. Bending, kneeling, or stooping	☐	☐	☐
9. Walking more than a mile	☐	☐	☐
10. Walking several blocks.	☐	☐	☐
11. Walking one block	☐	☐	☐
12. Bathing or dressing yourself	☐	☐	☐

SF-36 ™ Health Survey (continuation)

During the past 4 weeks, have you had any of the following problems with your work or other regular daily activities as a result of your physical health?

(Mark One Box on Each Line)

	Yes	No
13. Cut down the amount of time you spent on work or other activities	☐	☐
14. Accomplished less than you would like	☐	☐
15. Were limited in the kind of work or other activities	☐	☐
16. Had difficulty performing the work or other activities (for example, it took extra effort)	☐	☐

During the past 4 weeks, have you had any of the following problems with your work or other regular daily activities as a result of any emotional problems (such as feeling depressed or anxious)?

(Mark One Box on Each Line)

	Yes	No
17. Cut down the amount of time you spent on work or other activities . . .	☐	☐
18. Accomplished less than you would like	☐	☐
19. Didn't do work or other activities as carefully as usual	☐	☐

20. During the past 4 weeks, to what extent has your physical health or emotional problems interfered with your normal social activities with family, friends, neighbors, or groups? (mark one box)

☐ Not at all
☐ Slightly
☐ Moderately
☐ Quite a bit
☐ Extremely

21. How much bodily pain have you had during the past 4 weeks? (mark one box)

☐ None
☐ Very mild
☐ Mild
☐ Moderate
☐ Severe
☐ Very severe

22. During the past 4 weeks, how much did pain interfere with your normal work (including both work outside the home and housework)? (mark one box)

☐ Not at all
☐ A little bit
☐ Moderately
☐ Quite a bit
☐ Extremely

SF-36 ™ Health Survey (completion)

These questions are about how you feel and how things have been with you during the past 4 weeks. For each question, please give the one answer that comes closest to the way you have been feeling.

How much of the time during the past 4 weeks . . . . (Mark One Box on Each Line)

	All of the Time	Most of the Time	A Good Bit of the Time	Some of the Time	A Little of the Time	None of the Time
23. Did you feel full of pep?	☐	☐	☐	☐	☐	☐
24. Have you been a very nervous person?	☐	☐	☐	☐	☐	☐
25. Have you felt so down in the dumps that nothing could cheer you up? . .	☐	☐	☐	☐	☐	☐
26. Have you felt calm and peaceful? . . .	☐	☐	☐	☐	☐	☐
27. Did you have a lot of energy?	☐	☐	☐	☐	☐	☐
28. Have you felt downhearted and blue? .	☐	☐	☐	☐	☐	☐
29. Did you feel worn out?	☐	☐	☐	☐	☐	☐
30. Have you been a happy person?	☐	☐	☐	☐	☐	☐
31. Did you feel tired?	☐	☐	☐	☐	☐	☐

32. During the past 4 weeks, how much of the time has your physical health or emotional problems interfered with your social activities (like visiting with friends, relatives, etc.)? (mark one box)

☐ All of the time
☐ Most of the time
☐ Some of the time
☐ A little of the time
☐ None of the time

How TRUE or FALSE is <u>each</u> of the following statements for you.

(Mark One Box on Each Line)	Definitely True	Mostly True	Don't Know	Mostly False	Definitely False
33. I seem to get sick a little easier than other people	☐	☐	☐	☐	☐
33. I seem to get sick a little easier than other people	☐	☐	☐	☐	☐
34. I am as healthy as anybody I know	☐	☐	☐	☐	☐
35. I expect my health to get worse	☐	☐	☐	☐	☐
36. My health is excellent	☐	☐	☐	☐	☐

Signature of person filling out the form:
____

Date: __ __ /__ __ __ / __ __ __ __ (dd/mon/yyyy)

Appendix 4

AUDIT-C

The Alcohol Use Disorders Identification Test (AUDIT-C) is 3-item screen that can help identify patients who are hazardous drinkers or have active alcohol use disorders (including alcohol abuse or dependence). AUDIT-C is a modified version of the 10-question AUDIT instrument.

The AUDIT-C is scored on a scale of 0-12 (scores of 0 reflect no alcohol use). Each question has 5 choices. Points allotted are: a=0 points, b=1 point, c=2 points; d=3 points, e=4 points. In men, a score of 4 or more is considered positive; in women, a score of 3 or more is considered positive. Generally, the higher the AUDIT-C score, the more likely it is that the patient's drinking is affecting his/her health and safety.

1. How often do you have a drink containing alcohol? (a) Never [**Skip Qs 2-3**] (b) Monthly or less (c) 2 to 4 times a month (d) 2 to 3 times a week (e) 4 or more times a week	☐
2. How many standard drinks containing alcohol do you have on a typical day (when you are drinking)? (a) 1 or 2 (b) 3 or 4 (c) 5 or 6 (d) 7, 8, or 9 (e) 10 or more	☐
3. How often do you have six or more drinks on one occasion? (a) Never (b) Less than monthly (c) Monthly (d) Weekly (e) Daily or almost daily	☐
	Total Points: ☐

Appendix 5

Ancillary Study 1

Working Title: Cost Effectiveness Analysis of Metabolic Surgery for Obese Population with Type 2 Diabetes

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Possible Collaborators: All interested sites

Keywords: Bariatric surgery, metabolic surgery, cost-effectiveness analysis, type 2 diabetes, class I obesity

Primary Endpoint: Cost-effectiveness of metabolic surgery in overweight persons with type 2 diabetes as defined by ICER < $100,000 per QALY (cost utility analysis)

Introduction and Background:
There is a growing body of evidence suggesting that surgical treatment of type 2 diabetes with bariatric surgery is a safe and effective means of achieving durable improvement in glycemic control in those with severe obesity.[1,2] Thus, the concept of metabolic surgery is increasingly recognized and incorporated into a treatment algorithms for diabetes management [3-5]. Studies primarily focus on class II and III obese population, while class I obesity largely lacks randomized clinical data. Fortunately, many randomized control trials do include patients with BMI less than 35 as a part of their larger study, particularly in the ARMMS-T2D Collaborative. [6-10] However, many questions remain regarding the proper role of surgery in overall diabetes care in both obese and non-obese population. Economic evaluations are an important piece in this overall puzzle.

A recent joint statement by international diabetes organizations suggested surgery should be considered for patients with type 2 diabetes and class I obesity. [11] While there have been many cost-effectiveness studies published demonstrating the cost-effectiveness (not cost savings) of bariatric surgery in the morbidly obese population [12-15], these studies do not include class I obesity. A study published using gastric banding was published this year demonstrating cost-effectiveness with an incremental cost effectiveness ratio (ICER) of less than $50,000 per quality adjusted life year (QALY). [16] However, this was a study performed in Australia and converted to US dollars and only included gastric banding. There remains a lack of a robust cost effectiveness analysis in bariatric surgery based on randomized clinical trial data comparing surgery to intensive lifestyle and medical management for the class I obesity population in the United States.

The ARMMS-T2D Collaborative has one of the largest populations of randomized patients with class I obesity for study. By utilizing real world clinical data, a more accurate, applicable economic model can be developed.

Specific Aim:
- *Evaluate cost-effectiveness of metabolic surgery compared to intensive lifestyle medical interventions for all patients with BMI 30-35 with type 2 diabetes using a state-transition (Markov) microsimulation model.*

Methods and Study Design:
A cohort state-transition model (Markov model) was developed utilizing TreeAge Pro 2018 (TreeAge Software, Inc., Williamstown, MA) to compare cost-utility of two strategies: bariatric surgery vs. incentive lifestyle modifications.
- Cost effectiveness threshold will be set at $100,000/QALYs
    - Costs will be determined from the perspective of a third-party payer. No accounting for patient or caregiver time will be include.
    - Benefits will be reported using QALYs
- Prior studies have utilized BMI reduction for utility changes, but this study will utilize quality of life (QoL) surveys captured in respective trials as this population starts at a lower BMI which may under represent benefit to patients if used exclusively
- Background mortality will be estimated from US Third National Health and Nutritional Examination Survey[17]
- Costs will be determined from national references, such as Medicare fee schedule, referencing utilization of patients in the included studies rather than site specific costs due to significant geographic variation in costs of care
- Probabilistic sensitivity analyses will be performed to test assumptions made on model and impact on results

Timeline
Model development has already started and will continue to proceed without the required data. Once model development and testing has completed, data from all interested sites will need to be analyzed. Model development and debugging is expected to take 3 months and data analysis will take 1-2 months after model completion.

Role of ARMMS-T2D DCC
Staff members at DCC will need to compile and securely transmit required data from participating sites.

Data Analysis
Data provided by participating sites will be used to calculate the initial distribution patients such as preoperative BMI, age, and comorbidity status. Longitudinal clinical information will be used to estimate probabilities of developing certain outcomes, such as weight loss, comorbidity changes, and adverse events. The model will use the probabilities to calculate expected values of costs and QALYs for a similar patient population via a Markov microsimulation.

Data Required
- Identified information is not required and not requested. However, a patient identifier would be required to match and track patient data over time.
- Data required is limited to those patients with a BMI less than or equal to 35
- Basic patient demographics include pre-randomization information including age, gender, race/ethnicity, weight, BMI, presence or absence of comorbidities including Type 2 diabetes, hypertension, coronary artery disease, obstructive sleep apnea, gastroesophageal disease, osteoarthritis, or other comorbidities.
    - Medications required for comorbid conditions

    - Objective clinical data such as HgbA1C, fasting plasma glucose, lipid levels, blood pressure.
    - Duration of comorbidities prior to randomization
- Outcome information at longest recorded follow up including resolution of any of above comorbidities as well as changes or discontinuation of medications.
    - Objective clinical data such as HgbA1C, fasting plasma glucose, lipid levels, blood pressure.
- Adverse events and complications in both surgical and nonsurgical arms
- Healthcare utilization from surgery arm, if available
    - Preoperative work up
    - ED visits
    - Clinic visits
    - Medications
- Healthcare utilization of intensive lifestyle management/nonsurgical treatment arm, if available
    - Clinic visits
    - ED visits
    - Dietician visits
    - Medications
    - Exercise programs
- All quality of life surveys and types captured pre-operative and post-operative visits
- No direct cost information is required

Risk and Safety Concerns

Potential risks to ARMMS-T2D participants is expected to be minimal. The study involves de-identified data and the risk of harm is no greater than what would be encountered in daily life. No additional information will be collected from participants. Transmitted data will be stored within UPMC encrypted, cloud-based storage systems behind secure institutional firewalls with strong passwords. Data will not be stored on local devices.

There is no direct benefit for participants expected. No additional compensation will be provided.

Economic evaluations, particularly cost utility studies comparing US dollars per QALY earned allow for comparisons of interventions across multiple studies. Ultimately, information obtained from this study could support efforts to expand access to bariatric and metabolic surgery.

Impact on ARMMS-T2D Parent Study

No direct impact is expected on the parent study. The results of this study are expected to support any findings of benefits seen in surgical treatment of patients with obesity and type 2 diabetes within the parent study.

All interested sites are invited to participate. The only ARMMS-T2D resources required are the clinical data on randomized patients. The impact on the DCC is expected to include compilation and transmission of requested data in a secure fashion. Data analysis will occur within the TreeAge software or Microsoft Excel 2017.

Resources Required

TreeAge Pro 2018 will be used to carry out cost-effectiveness analysis. This software has been provided to this group without charge. Data storage will be provided by existing UPMC institutional infrastructure.

No additional resources are expected. No specimens are required. This study will not be undergoing an NIH peer-review process.

Disclosures

Dr. Mark Roberts performs consulting services for TreeAge Software, Inc. TreeAge Software, Inc. provides a free license of their software to Dr. Roberts for research use.

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